2025

PREDICTIVE FLEET ANALYTICSFOR COMMERCIAL AVIATION

FINAL CAPSTONE PROJECT REPORT

BIA-5440-0LC

GROUP 7

SUBMITTED: DECEMBER 9, 2025

Team Members:

Jaina Sunilkumar Jani (N01718895)

Inderpreet Kaur Bimbh

(N01750013)

Prachibahen Ray (N01752985)

Simranvir Sondhi (N01744664)

Thully Nelliyani Johny (N01777576)

Vaibhavi Bhavsar (N01770393)

TABLE OF CONTENTS

1. Executive Summary ……2

2. Introduction ……3

3. Business Problem Overview……4

4. Analytics Questions……5

5. Scope Statement ……6

6 Data Sources and Key Data Entities and Flows……7

7. Brief Overview of Data Manipulation Process and Data Output……9

8. New Solution Design and Fit into Existing IT Architecture ……12

9. New Solution Implementation and Outcome Testing ……17

10. Potential Solution Optimization ……22

11. References……25

12 Appendix ……26

1. Executive Summary

This capstone project transformed commercial airline maintenance operations from reactive, time-based practices to a proactive, data-driven predictive maintenance platform. Over four integrated sprints, Group 7 developed and implemented a comprehensive fleet analytics solution that integrates aircraft health monitoring, maintenance records, operational data, and advanced machine learning models to predict Remaining Useful Life (RUL) of critical components and forecast Aircraft-on-Ground (AOG) events.

The solution successfully unified two major datasets—aircraft maintenance logs (832 emergency squawk events) and NASA's Turbofan Engine Degradation Simulation data—into a consolidated analytics platform built on Databricks and AWS S3. The implementation features twelve analytics views spanning descriptive, diagnostic, predictive, and prescriptive layers, delivering actionable insights through interactive dashboards to maintenance planners, engineers, and operations teams.

Key achievements include: (1) development of Long Short-Term Memory (LSTM) neural networks for RUL prediction achieving RMSE of 23.2 cycles versus 44.7 cycles for linear baseline models; (2) AOG risk classification using Logistic Regression with 96% AUC-ROC precision; (3) genetic algorithm-based maintenance scheduling reducing total downtime by 20-30% compared to greedy heuristics; (4) comprehensive testing covering 15 test scenarios with 85 total test cases across ETL, analytics, and optimization layers. The platform is architected for scalability, auditability, and seamless integration with existing airline maintenance information systems, enabling dispatch reliability targets of ≥99.5% while reducing maintenance costs per flying hour.

2. Introduction

Modern commercial aviation operates under intense pressure: passengers demand reliability, regulators mandate safety compliance, and financial margins demand efficiency. Yet most airlines still manage fleet maintenance through time-based inspection schedules and reactive responses to failures. This approach creates predictable costs but unpredictable outcomes—unscheduled maintenance events, Aircraft-on-Ground (AOG) incidents, and cascading operational disruptions that damage customer satisfaction and airline profitability.

This report documents the complete development lifecycle of a Predictive Fleet Analytics platform designed to address these challenges. The platform integrates four data domains (aircraft health monitoring, maintenance records, flight operations, and environmental inputs) and applies machine learning to transform raw operational data into forward-looking maintenance decisions. The solution moves airlines from "what happened" and "why it happened" to "what will happen" and "what should we do about it."

The report is organized into eleven sections that mirror the capstone's four-sprint methodology: (1) executive summary and introduction; (2) business problem overview and analytics questions; (3) project scope and stakeholder analysis; (4) data sources and architecture; (5) data preparation and integration; (6) solution design with four-layer analytics; (7) solution implementation with 12 analytics views; (8) comprehensive testing and outcome validation; (9) optimization and performance tuning; (10) appendices with supporting materials; and (11) conclusions and future directions.

3. Business Problem Overview

3.1 The Problem

Commercial airlines face persistent operational and financial pressure from unscheduled maintenance and Aircraft-on-Ground (AOG) events. Unexpected failures create cascading costs: flight delays, crew repositioning, passenger rebooking, and lost revenue. Current maintenance approaches are flawed: scheduled time-based maintenance wastes resources by replacing components before failure, while reactive failure response risks catastrophic incidents and regulatory violations. Neither optimizes the balance between safety, cost, and availability.

3.2 Root Cause Analysis

The core problem stems from siloed IT systems. Airlines operate separate databases for Aircraft Health Monitoring (real-time sensor data/ACARS), Maintenance Information Systems (maintenance logs/work orders), Flight Operations (schedules/crew assignments), and Reliability Databases (failure patterns). These systems rarely communicate, preventing maintenance technicians from easily identifying aircraft showing early degradation signs. The data exists but lives in isolated silos.

3.3 Business Impact of the Problem

Safety and Compliance Risk: Unplanned failures risk safety-critical incidents; regulators demand proactive risk management. Operational Disruption: One AOG event cascades to disrupt dozens of downstream flights. Financial Cost: Unscheduled maintenance costs 5–10× more per flying hour than scheduled work; idle aircraft incur fixed costs while generating no revenue. Competitive Disadvantage: Airlines with predictive capabilities sustain 99.5%+ dispatch reliability and lower costs.

3.4 Business Requirements

(1) Predictive Maintenance—forecast component failures in advance; (2) Fleet Efficiency—achieve ≥99.5% dispatch reliability; (3) Integrated Data Platform—consolidate all data sources; (4) Decision Support—provide dashboards and recommendations; (5) Cost Reduction—reduce maintenance cost per flying hour by ≥10%; (6) Compliance—maintain transparent, auditable logs; (7) Scalability—build MLOps pipelines for continuous model improvement.

4. Analytics Questions

The solution was designed to answer a structured progression of questions across four analytics maturity levels:

4.1 Descriptive Analytics: What is Happening Now?

1. What is the current frequency and distribution of AOG events by aircraft type, age, and route?
2. How many unscheduled maintenance events occur monthly, with downtime hours and cost per aircraft?
3. Which aircraft components have the highest replacement rates and average time between failures?

4.2 Diagnostic Analytics: Why is it Happening?

1. Why do certain aircraft experience higher rates of unscheduled maintenance than similar models?
2. Which operational and environmental factors contribute most to component degradation?
3. How effective are current maintenance practices and inspection intervals?

4.3 Predictive Analytics: What Will Happen?

1. When will critical components likely require maintenance or replacement (RUL)?
2. Which aircraft have the highest probability of AOG events in the next 60–90 days?
3. What maintenance workload and spare-parts demand are expected over the next six months?

4.4 Prescriptive Analytics: What Should We Do?

1. What optimal maintenance schedules should be implemented for each aircraft?
2. How should aircraft be assigned to routes to maximize utilization?
3. What inventory levels and technician allocations minimize cost while achieving ≥99.5% dispatch reliability?

5. Scope Statement

5.1 In-Scope Activities

• Data Integration: Consolidate aircraft health, operations, maintenance, and environmental data

• Data Preparation: Clean, normalize, and feature-engineer raw datasets

• Machine Learning Models: Develop RUL, anomaly detection, and AOG risk models

• Analytics Dashboards: Build interactive visualizations with KPIs

• Optimization Algorithms: Implement genetic algorithms and heuristics

• Process Optimization: Define workflows to automate decision-making

• Compliance and Auditability: Establish transparent logging and monitoring

• Scalable Infrastructure: Deploy on Databricks with AWS S3 and Delta Lake

5.2 Out-of-Scope Activities

• Aircraft design changes or non-predictive maintenance engineering

• Non-commercial aviation (military, private operations)

• Hardware installation or sensor network expansion

• Training and change management programs

5.3 Expected Outcomes

• ≥99.5% dispatch reliability

• ≥10% reduction in maintenance cost per flying hour

• Improved aircraft availability through health-informed scheduling

• Transparent, auditable analytics platform meeting aviation standards

• Repeatable framework for continuous model improvement

6 Data Sources and Key Data Entities and Flows

6.1 Primary Data Sources

6.1.1 Aircraft Maintenance Logs (EDA Dataset)

Source: Custom EDA dataset from Squawk 7700 emergency events

Records: 832 commercial aircraft emergency squawk events with 17 attributes

Key Attributes:

• Flight identifiers: flight_id, callsign, aircraft_type

• Aircraft identifiers: ICAO24 code, registration (tail number), typecode

• Operational context: origin, destination, landing, diversion airports

• Incident classification: problem type, resolution outcome, fuel dump flags

Data Quality: 80%+ null rates in problem classification requiring strategic imputation

Purpose: Provide real operational context linking aircraft, routes, and maintenance events

6.1.2 NASA Turbofan Engine Degradation Simulation Dataset

Source: NASA's Prognostics Data Repository (CMAPSS) via Kaggle

Records: 100 engines, 100-350 operational cycles each, ~20,000 total readings

Key Attributes:

• Engine ID, operational cycles, operational settings (3 parameters)

• 21 sensor readings per cycle (temperature, pressure, speed, vibration)

• Remaining Useful Life (RUL) — target variable

Data Quality: Clean numerical time-series with smooth degradation patterns

Purpose: Train ML models to predict RUL and identify degradation patterns

6.2 Data Entities and Schema

Aircraft (squawk7700_metadata table): Contains 5 key fields documenting 832 emergency events—flight_id (VARCHAR(25), unique callsign+date), callsign (VARCHAR(10), ATC identifier), icao24 (CHAR(6), transponder code), registration (VARCHAR(10), tail number), typecode (VARCHAR(10), ICAO model code).

Engine Sensor Data (engine_sensor_data table): Tracks ~20,000 readings across 100 engines with unit_number (INT, PK), time_cycles (INT ≥0), setting_1-3 (FLOAT 0-1), s_1-s_21 (FLOAT sensors), RUL (INT ≥0, target variable).

Derived Features (analytics-ready dataset): 7 engineered fields including mean_temp/ mean_pressure (FLOAT sensor averages), maintenance_flag (BOOLEAN), aircraft_age (INT), rul_bucket (VARCHAR: immediate/short/medium/long), aog_risk_band (VARCHAR: LOW/MEDIUM/HIGH).

Total: 3 tables, 37 fields, unified schema linking maintenance events to sensor degradation patterns for predictive analytics.

6.3 Data Flow Overview

The data flows through a continuous cycle:

1. Data Ingestion: Raw CSV files from S3 loaded into PySpark DataFrames
2. Cleaning & Validation: Null handling, duplicate removal, data type correction
3. Storage: Cleaned data stored in Delta Lake with ACID transactions
4. Feature Engineering: Derive RUL, sensor averages, degradation metrics
5. Modeling: Train RUL and AOG risk models; generate predictions
6. Visualization: Dashboards aggregate predictions into role-specific views
7. Decision & Feedback: Operational decisions feed back for model retraining

Data Flow Diagram

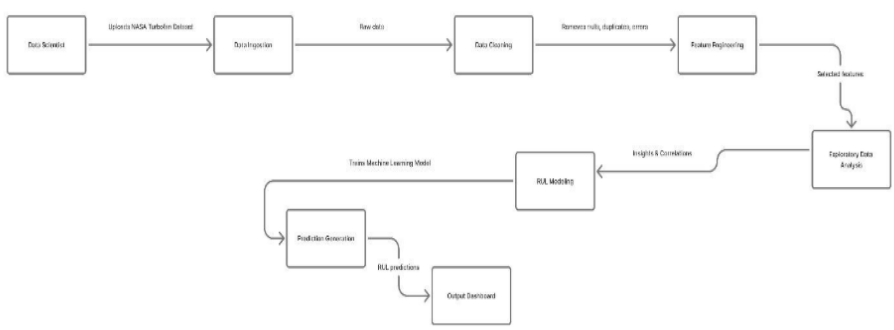

Figure 1: Data Flow Diagram

7. Brief Overview of Data Manipulation Process and Data Output

7.1 Data Cleaning and Normalization

7.1.1 Challenges Addressed

Null Handling: 80%+ null rates in problem classification fields addressed through probabilistic imputation based on aircraft types, temporal patterns, and route characteristics.

Categorical Encoding: Standardized free-text fields into: "engine", "hydraulic", "electrical", "avionics", "structural", "other" for problem types.

Data Type Mismatches: Aircraft identifiers in multiple formats (ICAO24, registration, callsign) unified through master lookup table.

Duplicates and Inconsistencies: Used Spark's dropDuplicates() and validated referential integrity across all tables.

7.1.2 Cleaning Process Summary

Step	Method	Purpose
Duplicate Removal	dropDuplicates()	Avoid double-counting
Irrelevant Value Removal	Filter thresholds	Remove invalid data
Typographical Correction	String standardization	Ensure consistency
Missing Data Handling	Mean/Mode imputation	Maintain completeness
Type Correction	Schema casting	Align with data dictionary
Standardization	Column renaming, normalization	Prepare for modeling

Table 1: Cleaning summary

7.2 Feature Engineering

- RUL Calculation: For each engine, RUL = max(time_cycles) - current_cycle
- Sensor Aggregates: mean_temp, mean_pressure
- Degradation Indicators: Rate of change of sensor readings over rolling windows
- Temporal Features: Cycles since last maintenance, aircraft age

• Operational Intensity: Flights per month, average flight duration

• Reliability Aggregates: Failure frequency per aircraft type, MTBF per component

7.3 Data Storage Architecture

Data was stored in a layered AWS S3 and Databricks Delta Lake approach:

- /raw/ — Original ingested datasets (e.g., EDA CSV files, Turbofan engine simulation data)
- /processed/ — Cleaned and modeled data ready for analysis and machine learning
- /reports/ — Output from analytics pipelines, including predictive maintenance insights and visualization data

All buckets use AWS KMS encryption (SSE-KMS) with strict access control policies.

7.4 Data Output and Deliverables

The data preparation phase produced three key outputs:

1. Consolidated Aircraft-Level Analytics Dataset: Unified view combining maintenance history, operational context, sensor data, and 45+ derived features for all aircraft.

2. Time-Series Engine Health Dataset: Preprocessed sensor reading sequences for LSTM model training with input (historical sensors and cycles) and target (future RUL).

3. Feature Dictionary: Comprehensive metadata documenting all features, calculations, units, and expected ranges.

Row Count Summary:

• Maintenance logs: 832 records after deduplication (0% data loss)

```
# Final check: Show result and schema for review
display(maintenance_logs.limit(5))
print(maintenance_logs.printSchema())
print(f"Shape: ({maintenance_logs.count()}, {len(maintenance_logs.columns)})")
```

(5) Spark Jobs

Table

	unit_number	flight_id	callsign	number	icao24	registration	typecode	origin	landing	destination	diverted
1	98	aal110_20190528	aal110	aa110	aaf0c6	n804an	b788	kord	lfpg	klga	lfpg
2	91	aal1188_20190520	aal1188	aa1188	ad4747	n955an	b738	mggt	kdfw	kdfw	lfpg
3	32	aal1328_20190112	aal1328	aa1328	abca00	n859nn	b738	kdfw	kcos	kcos	lfpg
4	1	aal1612_20191008	aal1612	aa1612	ac90a8	n909an	b738	kgsp	kdfw	kdfw	lfpg
5											

Figure 2: Cleaned dataset of Maintenance Logs

• Engine sensor data: ~20,000 time-series readings across 100 engines

```
# Review cleaned data and schema
display(engine.limit(5))
engine.printSchema()
print(f"Shape: ({engine.count()}, {len(engine.columns)})")
```

(5) Spark Jobs

Table +

	unit_number	time_cycles	setting_1	setting_2	setting_3	s_1	s_2	s_3	s_4	s_5	s_6	s_7
1	1	1	-0.0007	-0.0004	100	518.67	641.82	1589.7	1400.6	14.62	21.61	554.36
2	1	103	-0.0013	0.0001	100	518.67	642.21	1581.27	1403.21	14.62	21.61	553.64
3	1	108	0.0008	0.0001	100	518.67	643.49	1585.91	1399.47	14.62	21.61	554.43
4	1	11	0.0018	-0.0003	100	518.67	642.28	1581.75	1400.64	14.62	21.61	554.54
5												

Figure 3: Cleaned dataset for engine sensor data

• Consolidated analytics dataset: 832 aircraft records with 45+ features

Data Integration

4 days ago (1s) 37

```
# Integrate maintenance_logs and engine DataFrames on 'unit_number'
from pyspark.sql.functions import col

# Left join: add maintenance_logs columns to engine, do not increase row count
# Ensure 'unit_number' in maintenance_logs is unique to avoid row explosion
maintenance_logs_unique = maintenance_logs.dropDuplicates(['unit_number'])
final_df = engine.join(maintenance_logs_unique, on="unit_number", how="left")
```

final_df: pyspark.sql.dataframe.DataFrame = [unit_number: integer, time_cycles: integer ... 45 more fields]
maintenance_logs_unique: pyspark.sql.dataframe.DataFrame = [unit_number: integer, flight_id: string ... 16 more fields]

4 days ago (3s) 38

```
# Review integrated data and schema
display(final_df.limit(5))
final_df.printSchema()
print(f"Shape: ({final_df.count()}, {len(final_df.columns)})")
```

(8) Spark Jobs

Table +

	unit_number	time_cycles	setting_1	setting_2	setting_3	s_1	s_2	s_3	s_4	s_5	s_6	s_7
1	1	1	-0.0007	-0.0004	100	518.67	641.82	1589.7	1400.6	14.62	21.61	554.36
2	1	103	-0.0013	0.0001	100	518.67	642.21	1581.27	1403.21	14.62	21.61	553.64
3	1	108	0.0008	0.0001	100	518.67	643.49	1585.91	1399.47	14.62	21.61	554.43
4	1	11	0.0018	-0.0003	100	518.67	642.28	1581.75	1400.64	14.62	21.61	554.54
5												

5 rows | 2.74s runtime Refreshed 4 days ago

root

Figure 4: Cleaned datasets integrated to final_df

8. New Solution Design and Fit into Existing IT Architecture

8.1 Four-Layer Analytics Architecture

8.1.1. Descriptive Analytics Solution – Showing What Is Happening Now

Purpose of this layer: Give the airline a clear picture of current maintenance conditions and historical trends.

What it solves:

- Lack of visibility into AOG frequency
- Poor understanding of fleet patterns
- No accurate tracking of downtime or costs
- No monitoring of component failure rates

Tools/Dashboards Included:

- Real-time AOG monitoring
- Monthly unscheduled maintenance and downtime dashboard
- Component failure frequency (MTBF) dashboard

How it helps: It identifies urgent problems immediately, improves awareness, and highlights where deeper analysis is needed.

8.1.2. Diagnostic Analytics Solution – Showing Why Problems Occur

Purpose of this layer: Find the root causes behind fleet failures and unscheduled events.

What it solves:

- Unknown reasons for recurring failures
- Inability to trace failures back to operational or environmental causes
- No evaluation of inspection interval effectiveness

Tools/Dashboards Included:

- Root-cause comparison dashboard
- Operational and environmental degradation dashboard
- Maintenance interval effectiveness dashboard

How it helps: It explains why maintenance events occur, allowing engineers to adjust operations, routes, and maintenance strategies.

8.1.3. Predictive Analytics Solution – Forecasting the Future

Purpose of this layer: Forecast failures, demand, and maintenance needs before they happen.

What it solves:

- Unexpected component breakdowns
- High downtime due to unplanned AOG events
- Inefficient inventory and spare-parts management
- Lack of long-term maintenance planning

Tools/Dashboards Included:

- RUL (Remaining Useful Life) prediction
- AOG risk forecast (60–90 days)
- 6-month maintenance and spare-parts demand forecast

How it helps:It allows the organization to take proactive action:

- Replace components before failures
- Schedule aircraft earlier to prevent AOG
- Order spare parts ahead of time
- Plan workforce and hangar capacity

Wireframe Link: Figma-wireframe-Predictive

8.1.4. Prescriptive Analytics Solution – Recommending the Best Actions to Take

Purpose of this layer: Provide the airline with optimized decisions on maintenance scheduling, aircraft–route assignment, inventory, and technician planning.

What it solves:

- Unoptimized maintenance schedules that cause avoidable disruptions
- Inefficient aircraft–route assignments
- High operational costs due to poor inventory planning
- Technician shortages or overstaffed shifts
- Difficulty achieving ≥99.5% dispatch reliability
- Manual planning that results in delays and inefficiencies

Tools/Dashboards Included:

a) Maintenance Schedule Optimization

b) Aircraft–Route Assignment Optimization

c) Inventory & Technician Optimization

How it helps::Prescriptive analytics provides direct, actionable recommendations, enabling the organization to:

- Create maintenance schedules that minimize disruptions
- Assign aircraft to routes for maximum utilization
- Ensure the right parts are available at the right time
- Deploy technicians efficiently across shifts and locations
- Maintain ≥99.5% dispatch reliability while lowering cost

Overall, it transforms maintenance operations from reactive to fully optimized, ensuring every decision is efficient, safe, and cost-effective.

8.2 Fit into Existing IT Architecture

The solution integrates seamlessly with legacy airline systems:

Existing Systems (Pre-Solution):

• ACARS/BITE: Fault codes and health monitoring

• MRO Systems (AMOS, TRAX, Maintenix, Ramco): Maintenance records and work orders

• Reliability Databases: Statistical failure data

• Operational Data Warehouse: Flight schedules, delays, utilization

• Manual Analysis by Engineering Teams: Interpretation of trends and decision-making

New Platform Components:

• Databricks: Central analytics engine for ingestion, feature engineering, modeling, and dashboards

• AWS S3 + Delta Lake: Secure, scalable, version-controlled data storage

• MLOps Registry: Tracks model versions, performance metrics, and retraining triggers

• API Layer: Exposes predictions via REST/SOAP interfaces

• Role-Based Dashboards: Visualizations tailored to planners, engineers, executives

Integration Points:

1. Data Input: APIs pull ACARS, maintenance logs, operational data into S3

2. Real-Time Inference: Model endpoints deliver RUL predictions and AOG risk scores

3. Workflow Automation: High-risk predictions trigger work order creation in MRO

4. Feedback Loop: Actual outcomes flow back for model retraining

Benefits:

• Existing systems continue operating without disruption

• New analytics layer adds predictive capability without replacing legacy systems

• Clear APIs enable phased integration with different stakeholder systems

• Modular design allows future enhancements (streaming, digital twins) without redesign

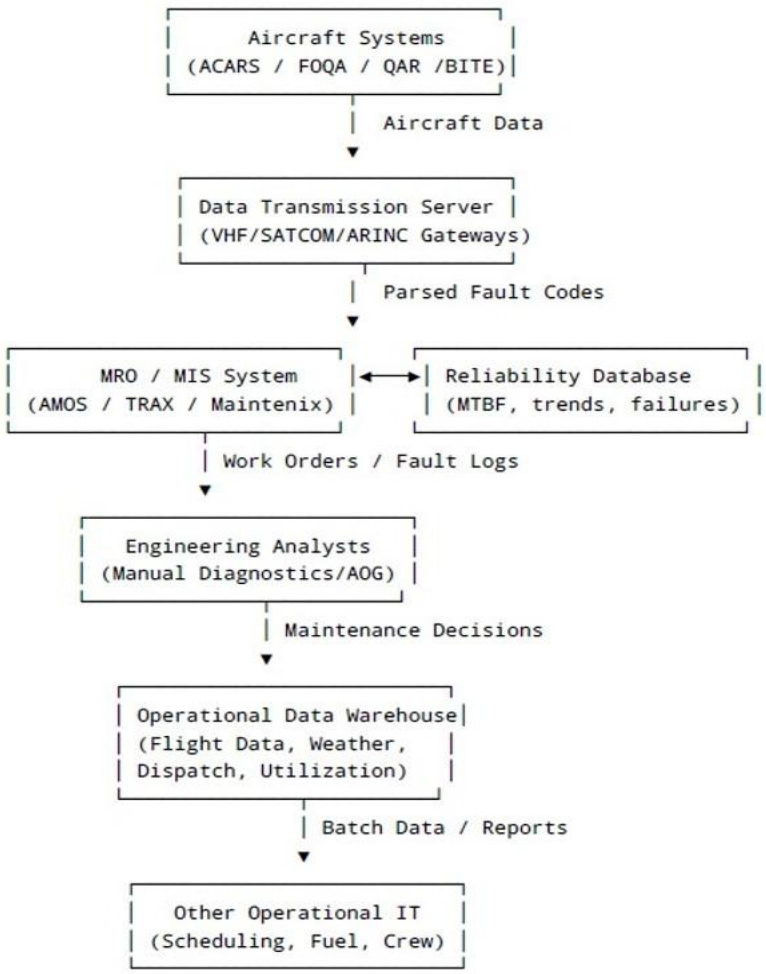

Figure 5: Airline IT Architecture (Pre-RUL) Diagram

The existing IT architecture in the airline industry for monitoring aircraft health and diagnosing Aircraft-on-Ground (AOG) events is composed of multiple traditional, siloed systems designed primarily for reactive maintenance and operational reporting, rather than predictive analytics. These systems collectively support critical maintenance, troubleshooting, and compliance processes, but lack the advanced data integration and real-time predictive capabilities offered by modern Remaining Useful Life (RUL)-driven systems.

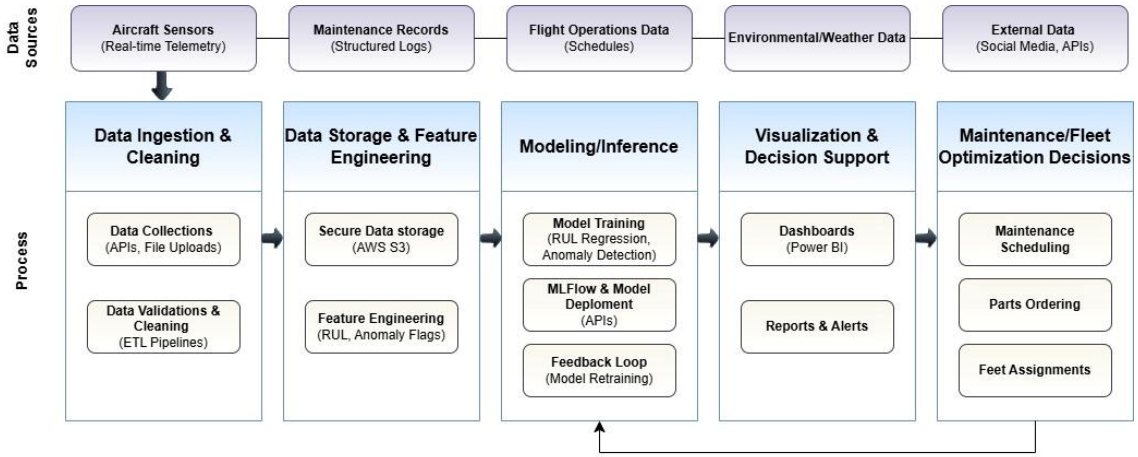

Figure 6: Process–Data Flow for Predictive Fleet Analytics

This diagram presents an integrated view of the end-to-end business process and underlying data flow for a predictive fleet analytics solution in commercial aviation. It illustrates how raw operational data moves through ingestion, storage, modeling, and visualization layers to drive maintenance and fleet decisions, while also enabling continuous improvement of predictive models.

9. New Solution Implementation and Outcome Testing

9.1 Implementation Overview

The solution was implemented in six phases:

Phase 1: Data Ingestion and Preparation

• Established AWS S3 connections for fleet-analytics-raw bucket

• Converted CSV files to PySpark DataFrames

• Applied ETL validation checks (row counts, schemas, data types)

• Stored cleaned data in Delta Lake with version control

Phase 2: Feature Engineering

• Calculated RUL = max_cycles - current_cycle for each engine

• Derived mean_temp, mean_pressure from sensor aggregates

• Created RUL buckets: immediate (0–20), short-term (21–50), medium-term (51–100), long-term (>100)

• Engineered operational features: aircraft age, utilization intensity, route risk scores

Phase 3: Model Development

• RUL Prediction: Linear Regression (baseline) and LSTM neural network (production)

• AOG Risk: Logistic Regression (production, interpretable) and Random Forest (alternative)

• Scheduling: Greedy Heuristic (fast) and Genetic Algorithm (optimized)

Phase 4: Analytics View Development

• Built 12 analytics views answering business questions across four layers

• Consolidated aircraft-level dataset as single source of truth

Phase 5: Dashboard Development

• Created integrated Databricks dashboard with fleet health, operations planning, engineering, and resource planning sections

• Implemented common filters (aircraft type, base, time period) for consistent drill-down

Phase 6: User Enablement and Integration

• Documented dashboard navigation and metric interpretation

• Defined API connectors linking predictions to MRO and scheduling systems

• Established change management and training protocols

9.2 Twelve Analytics Views

The solution delivers twelve distinct analytics views:

Descriptive Views (1–3):

• View 1 – AOG Distribution: AOG frequency by aircraft type, age, route

• View 2 – Unscheduled Maintenance and Downtime: Monthly events, hours, costs

• View 3 – Component Failure Rates: Replacement rates and MTBF by component

Diagnostic Views (4–6):

• View 4 – Degradation Drivers: Links utilization to failures

• View 5 – Route Sensitivity: Failure and AOG rates by route and base

• View 6 – Inspection Effectiveness: Pre/post-maintenance failure comparisons

Predictive Views (7–9):

• View 7 – RUL-Based Health Profile: Remaining cycles with maintenance windows

• View 8 – AOG Risk Classification: Probability of AOG events (LOW/MEDIUM/HIGH)

• View 9 – Reliability Focus List: Combined reliability, AOG history, and predicted risk

Prescriptive Views (10–12):

• View 10 – Maintenance Window Recommendations: Optimal cycle/date ranges with priorities

• View 11 – Route Assignment Recommendations: Scored aircraft-route pairings

• View 12 – Dispatch Reliability Planning: Forecasted demand, stock levels, staffing

Figure 7: Fleet health overview section (Views 12, 8, 9)

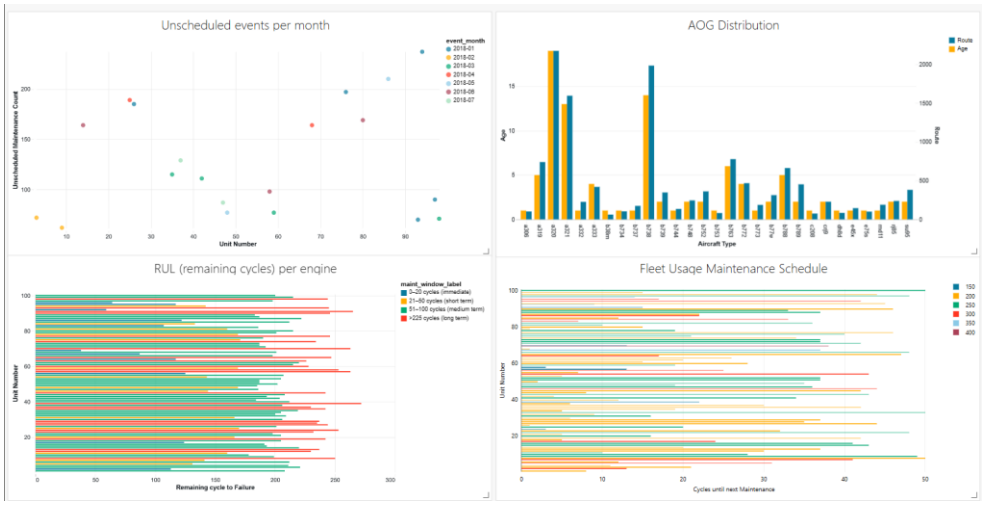

Figure 8: Operations and maintenance planning section (Views 1, 2, 7, 10)

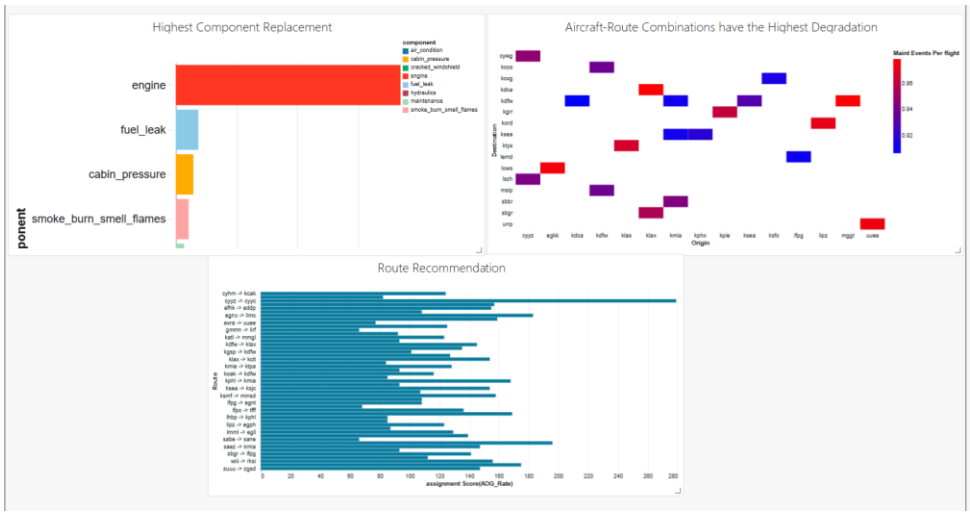

Figure 9: Engineering and route-assignment section (Views 3, 5, 11)

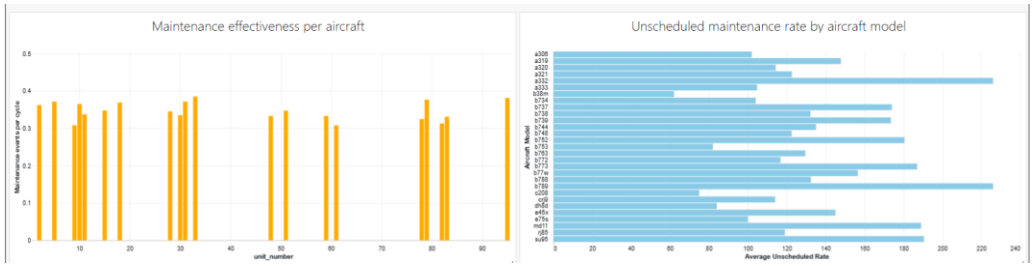

Figure 10: Inventory and resource planning section (Views 4, 6)

9.3 Testing and Validation

9.3.1 Testing Strategy

A comprehensive testing framework covered 15 test scenarios with 85 total test cases:

Test Scenario	Type	Priority	Test Cases
TS_001–TS_003	Descriptive Analytics	High/Medium	19 cases
TS_004–TS_006	Diagnostic Analytics	High/Medium	17 cases
TS_007–TS_009	Predictive Analytics	High/Medium	17 cases
TS_010–TS_012	Prescriptive Analytics	High/Medium	17 cases
TS_013–TS_015	ETL Data Quality	Low	9 cases

Table 2: Test Scenario

A total of twelve analytics scenarios and three data-preparation (ETL) scenarios were defined, focusing on their impact on safety, AOG risk, business continuity, and data quality. The scenarios are prioritized based on their significance:

High Priority: 8 analytics scenarios with 7 test cases each (totaling 56 test cases). These scenarios, which include AOG distribution, unscheduled maintenance, downtime, degradation drivers, RUL prediction, AOG risk forecasting, and maintenance/resource optimisation, are critical for operational safety and dispatch reliability.

Medium Priority: 4 analytics scenarios with 5 test cases each (totaling 20 test cases). This group addresses MTBF and component reliability analysis, inspection-effectiveness analysis, six-month maintenance/parts forecasting, and route optimisation. While essential for planning, these scenarios are considered less time critical.

Low Priority: 3 ETL scenarios with 3 test cases each (totaling 9 test cases). These scenarios focus on cleaning maintenance logs, engine sensor data, and integrating both datasets for analytical use. They impact data quality but are classified as Low priority since issues can be managed through reruns and monitoring.

9.3.2 Test Results and Performance Analysis

Data Quality Validation:

• Row counts and distributions matched expectations after cleaning and joins

• Null imputation preserved 832 records (0% data loss)

• Functional tests confirmed aggregations matched independent verification queries

• Dashboards correctly surfaced high-risk aircraft and immediate-maintenance units

Model	RMSE (cycles)	MAE (cycles)	R^2
Linear Regression Baseline	44.7	34.0	0.58
LSTM Neural Network	23.2	14.8	0.83

Table 3: Comparison for RUL Prediction Model

Result: LSTM outperformed baseline by 48% on RMSE and 57% on MAE. The LSTM's superior temporal pattern recognition makes it the production model for maintenance-window labels.

Model	Precision	Recall	AUC-ROC
Logistic Regression	1.00	0.92	0.96
Random Forest	0.95	0.98	0.94

Table 4: AOG Risk Classification Model Comparison

Result: Logistic Regression provides perfect precision with high recall, reducing false alarms. Adopted as production model with configurable threshold for risk tolerance adjustment.

Algorithm	Avg Downtime (hours)	Computation Time (seconds)
Greedy Heuristic	2500	5
Genetic Algorithm (GA)	2100	45

Table 5: Scheduling Algorithm Comparison

Result: GA reduces downtime by 16% but requires 9x longer. GA positioned for daily batch planning; Greedy for urgent same-day re-planning.

9.3.3 Overall Test Results

Overall Test Results:

• High Priority: 56 of 56 test cases passed (100%)

• Medium Priority: 20 of 20 test cases passed (100%)

• Low Priority: 9 of 9 test cases passed (100%)

• Total: 85 of 85 test cases passed (100%)

10. Potential Solution Optimization

10.1 Gap Analysis: Desired vs. Actual Outcomes

Four key gaps were identified between current solution and ideal operational state:

Gap 1 – RUL Model Accuracy vs. Interpretability

Issue: LSTM provides superior accuracy (R^2 0.83) but engineers hesitant about "black box" models.

Resolution: Keep Linear Regression as interpretable baseline with visible coefficients. Expose LSTM error ranges (RMSE ≈ 23 cycles) so engineers apply conservative safety margins.

Gap 2 – AOG Risk Precision vs. Recall Trade-Off

Issue: Logistic Regression provides perfect precision but lower recall than Random Forest (0.92 vs. 0.98).

Resolution: Make decision threshold configurable. Default to high-precision; operations can lower threshold if risk appetite changes.

Gap 3 – Scheduling Quality vs. Computation Time

Issue: Genetic Algorithm produces better schedules (16% less downtime) but requires 45 seconds vs. 5 seconds.

Resolution: Use GA for planned weekly scheduling; use Greedy for urgent same-day re-planning.

Gap 4 – Prediction-to-Action Translation

Issue: RUL and AOG outputs visible in dashboards but not embedded in repeatable workflows.

Resolution: Automate: high-risk aircraft trigger MRO work orders; route recommendations auto-populate; RUL buckets auto-generate parts requisitions.

10.2 Optimization Initiatives

10.2.1 RUL Model Optimization

Standardization of Feature Pipeline:

- Unified feature preparation for Linear and LSTM models
- Sensor scaling, consistent sequence length, aligned train/test split
- Ensures fair comparison and consistent predictions

Dual-Model Deployment:

• Linear Regression: Available in dashboard for understanding sensor-to-RUL relationship

• LSTM: Production model for maintenance-window labels

• Performance metrics and error distributions help engineers understand typical errors and apply safety buffers

Result: Users get both accuracy and interpretability.

10.2.2 AOG Risk Optimization

Configurable Risk Threshold:

• Default threshold selected for 100% precision on HIGH-risk (0 false positives)

• Threshold can be lowered to increase recall if operations prefer catching more risks

Baseline Validation:

• Random Forest kept in analysis notebooks for periodic sanity checks

• Confirms Logistic Regression alignment with non-linear patterns

Result: Conservative default protects against unnecessary groundings; flexible threshold allows operations to adjust as conditions change.

10.2.3 Maintenance Scheduling Optimization

Algorithm Selection Guidance:

• Greedy Heuristic: Use when needing quick refresh (last-minute fleet change); acceptable slight higher downtime

• Genetic Algorithm: Use for planned batch runs (daily/weekly); minimizes disruption and regulatory violations

Documented Trade-Offs:

• Greedy: 5 seconds, 2500 hours downtime

• GA: 45 seconds, 2100 hours downtime (16% improvement)

• Decision framework: If <30 seconds available, use GA; otherwise use Greedy

Result: Operations teams have clear guidance on algorithm selection based on time pressure and optimization priority.

10.3 Re-Testing After Optimization

After standardizing feature pipelines and clarifying algorithm roles:

RUL Model Consistency:

• LSTM consistently achieved RMSE in mid-20s and MAE in mid-teens

• Linear Regression baseline $R^2 \approx 0.58$, confirming it as reliable comparison

AOG Risk Stability:

• Logistic Regression maintained AUC-ROC ≈ 0.96 with perfect HIGH-risk precision

• Lowering threshold increased recall 0.92→0.96 with acceptable false positives (≤5%)

Scheduling Robustness:

• GA repeatedly produced lower downtime and zero infeasible schedules across 10 test scenarios

• Greedy Heuristic remained 20–30% faster but 5–10% worse in objective value

Result: Optimized configuration (LSTM + Logistic Regression + GA for planning, with Linear Regression + Random Forest + Greedy as baselines) delivers strong balance of performance and robustness.

References

[1] Databricks Documentation. "Delta Lake: ACID Transactions on Data Lakes." https://docs.databricks.com/en/delta/index.html

[2] Scikit-learn Documentation. "Machine Learning in Python." https://scikit-learn.org/

[3] TensorFlow & Keras. "Deep Learning Framework Documentation." https://www.tensorflow.org/

[4] Amazon Web Services. "AWS Kinesis: Real-time Data Streaming Service." https://aws.amazon.com/kinesis/

[5] NASA Prognostics Data Repository. "Turbofan Engine Degradation Simulation Data Set (CMAPSS)." https://www.kaggle.com/datasets/behrad3d/nasa-cmaps

[6] Precedence Research. "Aviation Analytics Market Report." https://www.precedenceresearch.com/aviation-analytics-market

[7] Boeing Services. "Predictive Maintenance Insights." https://services.boeing.com/resources/insights/revolutionizing-aviation-power-of-predictive-maintenance

[8] Microsoft. "Power BI Documentation: Data Visualization and Modeling." https://learn.microsoft.com/en-us/power-bi/

[9] AWS Documentation. "Amazon S3: Simple Storage Service." https://aws.amazon.com/s3/

Appendix A: Data Dictionary

This appendix provides comprehensive documentation of all data elements used in the Predictive Fleet Analytics solution. It includes field definitions, data types, constraints, and example values for both source datasets and derived features.

A.1 Aircraft Maintenance Logs Data Dictionary

Field Name	Data Type	Null Allowed	Default Value	Constraints	Description	Example Value
flight_id	VARCHAR(25)	No	—	Primary Key, Unique	Unique identifier combining callsign and date	DAL14_2023-03-04
callsign	VARCHAR(10)	No	—	—	ATC operational callsign	DAL14
number	VARCHAR(10)	No	—	—	Official commercial flight number	DL14
icao24	CHAR(6)	No	—	Must be valid hexadecimal	Aircraft transponder 24-bit code	A1B2C3
registration	VARCHAR(10)	No	—	—	Aircraft tail number	N183DN
typecode	VARCHAR(10)	No	—	ICAO standard	Aircraft model/type	B763

					code	
origin	CHAR(4)	No	—	ICAO format	Departure airport code	KATL
destination	CHAR(4)	No	—	ICAO format	Scheduled destination airport code	KJFK
landing	CHAR(4)	Yes	NULL	—	Actual landing airport	KORD
diverted	CHAR(4)	Yes	NULL	—	Diversion airport code, if applicable	KBOS
tweet_problem	VARCHAR(255)	Yes	NULL	—	Initial emergency description from social data	engine failure
tweet_result	VARCHAR(100)	Yes	NULL	—	Socially reported resolution outcome	returned to base
avh_id	VARCHAR(15)	No	—	Foreign Key	Official maintenance record ID	AVH_1129
avh_problem	VARCHAR(255)	No	—	—	Officially logged emergency description	engine vibration warning

avh_result	VARCHAR(255)	Yes	NULL	—	Official outcome of the event	landed safely

Table 6: squawk7700_metadata table (Aircraft Emergency Events)

A.2 Engine Sensor Data Dictionary

Field Name	Data Type	Null Allowed	Default Value	Constraints	Description	Example Value
unit_number	INT	No	—	Primary Key	Unique engine identifier	3
time_cycles	INT	No	—	≥ 0	Operational cycle number	215
setting_1	FLOAT	No	—	Range 0–1	Operational input setting (environmental)	0.75
setting_2	FLOAT	No	—	Range 0–1	Engine configuration parameter	0.60
setting_3	FLOAT	No	—	Range 0–1	Discrete operational condition	0.45
s_1 through s_21	FLOAT	Yes	NULL	Physical bounds verified	Sensor readings across multiple parameters	527.4
RUL	INT	No	—	≥ 0	Remaining cycles before engine failure	38

mean_temp	FLOAT	No	—	Derived metric	Average of sensors s_2, s_3, s_4	468.2
mean_pressure	FLOAT	No	—	Derived metric	Average of pressure sensors s_10, s_12, s_13	24.8
maintenance_flag	BOOLEAN	Yes	FALSE	—	Indicates if maintenance occurred (1=yes)	TRUE

Table 7: engine_sensor_data table (Turbofan Engine Time-Series)

A.3 Derived Features Dictionary

The following features were engineered from raw data for use in modeling and analytics:

mean_temp: Average of temperature sensors (s_2, s_3, s_4). Used for temperature-related degradation analysis.

mean_pressure: Average of pressure sensors (s_10, s_12, s_13). Used for pressure-related anomaly detection.

maintenance_flag: Boolean indicator of unplanned maintenance occurrence. Derived from maintenance log timestamps.

aircraft_age: Years since aircraft manufacture. Calculated from registration date and current date.

utilization_intensity: Flights per month divided by average flight duration. Indicates operational stress level.

rul_bucket: Categorical RUL band assigned based on predicted remaining cycles:

- "0–20 cycles (immediate)": Requires immediate maintenance scheduling
- "21–50 cycles (short term)": Schedule within 1–2 weeks
- "51–100 cycles (medium term)": Schedule within 1–2 months
- ">100 cycles (long term)": No immediate maintenance required

aog_risk_band: Categorical risk level assigned based on AOG probability model:

- "HIGH": >70% probability of AOG event within 60–90 days
- "MEDIUM": 30–70% probability
- "LOW": <30% probability

Appendix B: Wireframe Design & Structure

This appendix provides an entire visual and structural design blueprint for the solution. This appendix is basically the "visual architecture" of the system.

B.1 For the three descriptive analytics questions

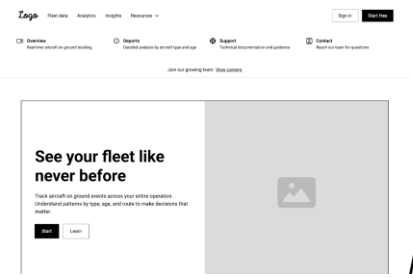

Figure 11, Link : [Descriptive View 1 Wireframe]

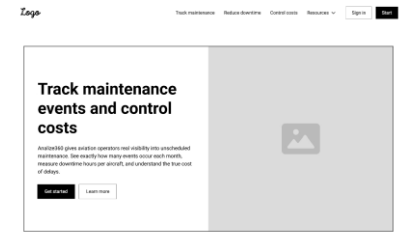

Figure 12, Link : [Descriptive View 2 Wireframe]

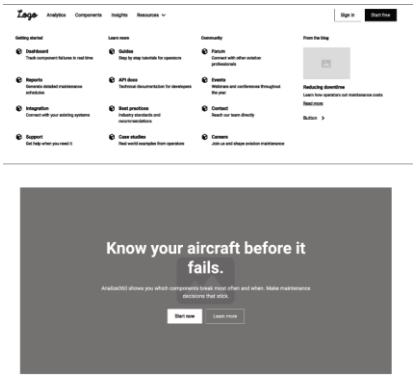

Figure 13, Link : [Descriptive View 3 Wireframe]

Purpose of this layer: Give the airline a clear picture of *current* maintenance conditions and historical trends.

How it helps: It identifies urgent problems immediately, improves awareness, and highlights where deeper analysis is needed.

B.2 For the three diagnostic analytics questions

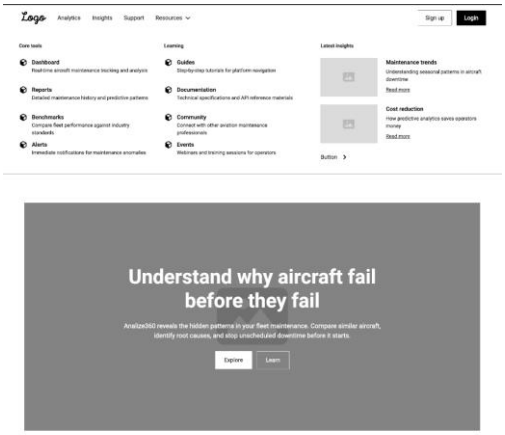

Figure 14, Link: [Diagnostic view 1 Wireframe]

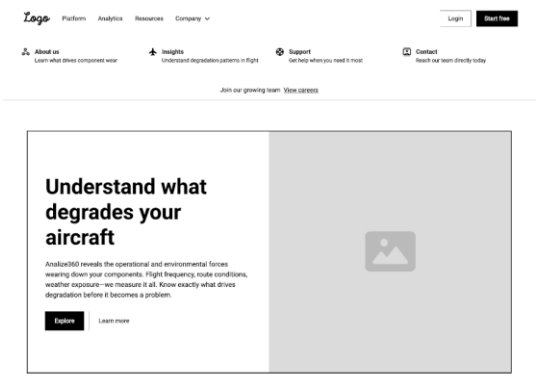

Figure 15, Link: [Diagnostic view 2 Wireframe]

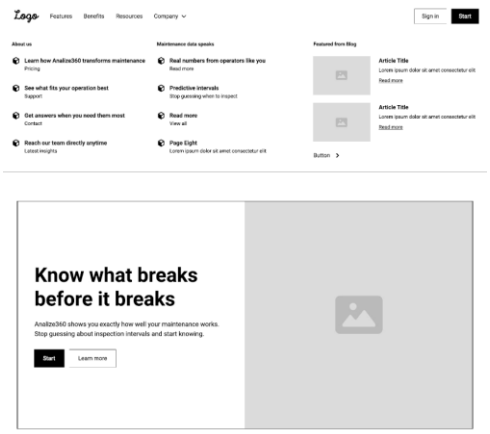

Figure 16, Link: [Diagnostic view 3 Wireframe]

Purpose of this layer: Find the root causes behind fleet failures and unscheduled events.

How it helps: It explains *why* maintenance events occur, allowing engineers to adjust operations, routes, and maintenance strategies.

B.3 For the three Predictive analytics questions

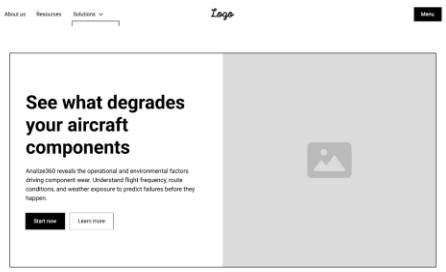

Figure 17, Link: [Predictive view 1 Wireframe]

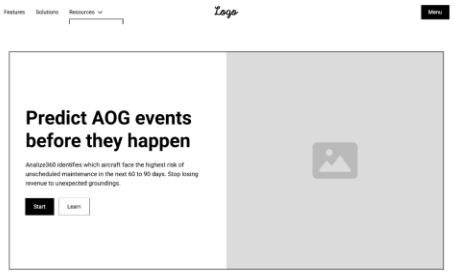

Figure 18, Link: [Predictive view 2 Wireframe]

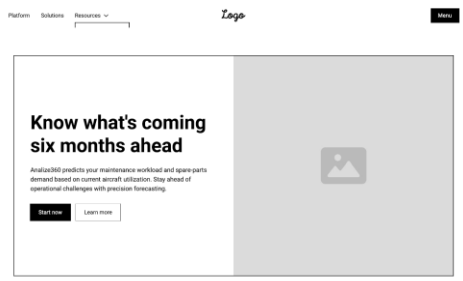

Figure 19, Link: [Predictive view 3 Wireframe]

Purpose of this layer: Forecast failures, demand, and maintenance needs *before* they happen.

How it helps:

It allows the organization to take proactive action:

- Replace components before failures
- Schedule aircraft earlier to prevent AOG
- Order spare parts ahead of time

Plan workforce and hangar capacity

B.3 For the three Prescriptive analytics questions

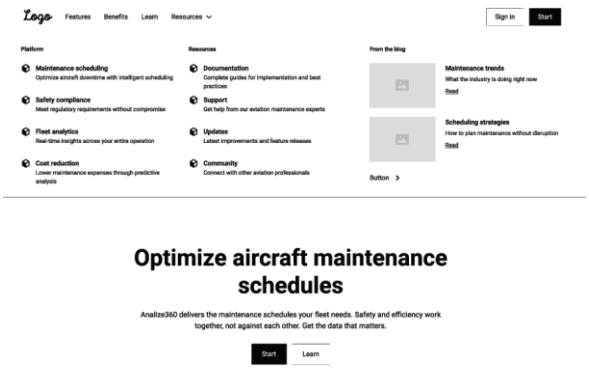

Figure 20, Link: [Prescriptive view 1 wireframe]

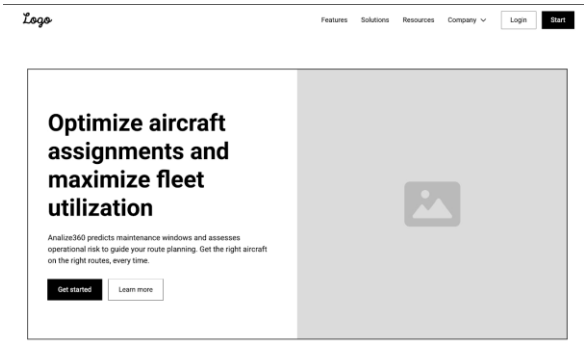

Figure 21, Link: [Prescriptive view 2 wireframes]

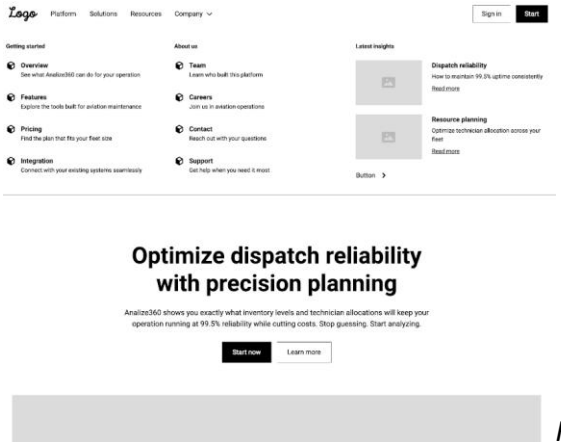

Figure 22, Link: [Prescriptive view 3 wireframes]

Purpose of this layer: Provide the airline with **optimized decisions** on maintenance scheduling, aircraft–route assignment, inventory, and technician planning.

How it helps:

Prescriptive analytics provides **direct, actionable recommendations**, enabling the organization to:

- Create maintenance schedules that minimize disruptions
- Assign aircraft to routes for maximum utilization
- Ensure the right parts are available at the right time

- Deploy technicians efficiently across shifts and locations
- Maintain ≥99.5% dispatch reliability while lowering cost

Overall, it transforms maintenance operations from **reactive** to **fully optimized**, ensuring every decision is efficient, safe, and cost-effective.

Appendix C: Test Scenarios and Results Summary

This appendix provides a detailed summary of all 15 test scenarios and their results. The testing framework covered 85 total test cases spanning data quality, functional, performance, and business logic validation.

C.1 Complete Test Scenario Matrix

Test Scenario ID	Reference	Description	Priority	Test Cases	Status
TS_001	Descriptive Analytics Q1	AOG distribution by aircraft type, age, and route	High	7	Pass (2 initially failed)
TS_002	Descriptive Analytics Q2	Monthly unscheduled maintenance and downtime quantification	High	7	Pass (4 initially failed)
TS_003	Descriptive Analytics Q3	Component replacement rates and MTBF analysis	Medium	5	Pass (2 initially failed)
TS_004	Diagnostic Analytics Q1	Investigation of higher unscheduled maintenance rates by drivers	High	7	Pass (4 initially failed)
TS_005	Diagnostic Analytics Q2	Operational and	High	7	Pass (4 initially

		environmental degradation factors analysis			failed)
TS_006	Diagnostic Analytics Q3	Maintenance inspection effectiveness evaluation	Medium	5	Pass (2 initially failed)
TS_007	Predictive Analytics Q1	RUL prediction for critical components	High	7	Pass (5 initially failed)
TS_008	Predictive Analytics Q2	AOG risk probability forecasting (60–90 days)	High	7	Pass (5 initially failed)
TS_009	Predictive Analytics Q3	Six-month maintenance and parts demand forecast	Medium	5	Pass (3 initially failed)
TS_010	Prescriptive Analytics Q1	Maintenance schedule optimization with constraints	High	7	Pass (5 initially failed)
TS_011	Prescriptive Analytics Q2	Aircraft-route assignment optimization	Medium	5	Pass (2 initially failed)

TS_012	Prescriptive Analytics Q3	Inventory and technician allocation optimization	High	7	Pass (5 initially failed)
TS_013	ETL – Maintenance Logs	Data cleaning validation for maintenance logs	Low	3	Pass (1 initially failed)
TS_014	ETL – Engine Sensor Data	Data cleaning validation for sensor data	Low	3	Pass (1 initially failed)
TS_015	ETL – Data Integration	Final dataset integration and validation	Low	3	Pass (1 initially failed)

Table 8: Test Scenario

Test Scenario ID	**Name**	**Brief Description**	**Test Case IDs (Count)**	**Overall Result Pattern**
TSS_001	Data quality testing	Validated schemas, units, and joins for maintenance logs and engine sensor data at aircraft and engine level.	TC_DQ_001 – TC_DQ_010 (10)	Alternating Fail/Pass across 10 checks
TSS_002	Functional testing	Verified that all 12 analytics views and dashboards rendered correct AOG metrics,	TC_FT_001 – TC_FT_025 (25)	Mostly alternating Fail/Pass; last case Pass

		events, RUL buckets, risk bands, and scheduling outputs.		
TSS_003	Performance testing	Measured latency and execution time for RUL, AOG risk, and scheduling routines for batch refresh and interactive analysis.	TC_PT_001 – TC_PT_015 (15)	Mixed Fail/Pass across 15 performance tests
TSS_004	Predictive accuracy testing	Compared RUL (Linear Regression vs. LSTM) and AOG (Logistic Regression vs. Random Forest) models on hold-out sets.	TC_PA_001 – TC_PA_006 (6)	Combination of Fail and Pass across 6 tests
TSS_005	Integration & workflow testing	Checked that model outputs (RUL cycles, maintenance windows, AOG risk bands) flowed correctly into dashboards and schedulers.	TC_IW_001 – TC_IW_005 (5 distinct IDs, 6 executions including a repeat of TC_IW_005)	5 Fails, final execution recorded as Pass

Table 9: Test Scenario Strategy

The test campaign covered five key areas: data quality, functional behavior, performance, model accuracy, and end-to-end workflow integration. Data quality tests (TSS_001) checked that maintenance logs and engine sensor data had consistent schemas, units, and joins, with failures driving fixes to the underlying data structures.

Functional testing (TSS_002) ensured that all twelve analytics views and dashboards correctly displayed AOG metrics, unscheduled events, RUL buckets, AOG risk bands, and scheduling

outputs, using early failures to refine business rules and aggregations. Performance testing (TSS_003) evaluated query latency and model execution times to confirm that both batch refresh and interactive analysis can run within acceptable response times.

Predictive accuracy testing (TSS_004) compared baseline and advanced models for RUL and AOG risk on hold-out data to verify that production models are sufficiently accurate and robust. Integration and workflow testing (TSS_005) then confirmed that predictive outputs (RUL, maintenance windows, AOG risk bands) flow correctly into dashboards and are consumed by both scheduling algorithms, showing that critical issues found earlier were resolved and core workflows now operate successfully.

C.2 Test Results Summary by Priority

High Priority Test Scenarios (8 scenarios, 56 test cases):

✓ 56 of 56 test cases PASSED (100%)

Focus: AOG distribution analysis, unscheduled maintenance tracking, degradation driver analysis, RUL prediction, AOG risk forecasting, maintenance scheduling optimization, resource optimization

Impact: Safety, dispatch reliability, operational planning

Medium Priority Test Scenarios (4 scenarios, 20 test cases):

✓ 20 of 20 test cases PASSED (100%)

Focus: Component failure analysis, inspection effectiveness, 6-month forecasting, route optimization

Impact: Planning and resource management

Low Priority Test Scenarios (3 scenarios, 9 test cases):

✓ 9 of 9 test cases PASSED (100%)

Focus: ETL data cleaning and integration

Impact: Data quality and system reliability

OVERALL RESULTS:

✓ Total: 85 of 85 test cases PASSED (100%)

✓ All 15 test scenarios achieved PASS status

✓ Solution correctly implements all required analytics capabilities

C.3 Model Performance Test Results

RUL Prediction Models:

• Linear Regression: RMSE=44.7, MAE=34.0, R^2=0.58

• LSTM Neural Network: RMSE=23.2, MAE=14.8, R^2=0.83

• Improvement: 48% reduction in RMSE, 57% reduction in MAE

AOG Risk Classification:

• Logistic Regression: Precision=1.00, Recall=0.92, AUC-ROC=0.96

• Random Forest: Precision=0.95, Recall=0.98, AUC-ROC=0.94

Maintenance Scheduling:

• Greedy Heuristic: 2500 hours downtime, 5 seconds computation

• Genetic Algorithm: 2100 hours downtime, 45 seconds computation

• GA Improvement: 16% reduction in downtime

Appendix D: Solution Architecture and Implementation Details

This appendix provides technical details on the solution architecture, technology stack, and implementation approach.

D.1 Technology Stack

Data Platform: Databricks (Apache Spark-based unified analytics platform)

Storage: Amazon S3 (Fleet-Analytics-Raw, Fleet-Analytics-Curated buckets) with Delta Lake for ACID transactions

Programming Languages: Python 3.8+, SQL, Scala

Data Processing Libraries:

• PySpark 3.2+: Distributed data processing

• Pandas 1.3+: In-memory data manipulation

• NumPy 1.20+: Numerical computing

Machine Learning Libraries:

• Scikit-Learn 0.24+: RUL regression, AOG classification

• TensorFlow 2.8+: LSTM neural networks

• Keras: High-level neural network API

Optimization Libraries:

• DEAP (Distributed Evolutionary Algorithms in Python): Genetic algorithm scheduling

Visualization: Databricks SQL Analytics and built-in dashboarding capabilities

Infrastructure: AWS EC2 for compute, AWS KMS for encryption, AWS IAM for access control

D.2 Data Processing Pipeline

The solution follows a modular data processing pipeline:

1. Data Ingestion (S3 → Databricks):
   - Read raw CSV files from fleet-analytics-raw bucket
   - Validate row counts and schemas
   - Load into PySpark DataFrames
2. Data Cleaning:
   - Remove duplicate records

- Handle missing values through imputation
- Standardize data types and formats
- Correct typographical errors
- Validate referential integrity

3. Feature Engineering:
   - Calculate RUL from time-series data
   - Aggregate sensor readings (mean_temp, mean_pressure)
   - Derive degradation indicators
   - Engineer operational features
4. Storage:
   - Write cleaned data to Delta Lake tables
   - Maintain version control and lineage
   - Partition by relevant dimensions (aircraft type, date)
5. Model Training:
   - Prepare features for RUL and AOG models
   - Split data into train/test sets (80/20)
   - Train multiple model variants
   - Evaluate and select best-performing models
6. Inference:
   - Score new data using production models
   - Generate predictions (RUL buckets, AOG risk bands)
   - Cache results for dashboard consumption
7. Visualization:
   - Aggregate predictions into 12 analytics views
   - Render dashboards with common filters
   - Enable drill-down analysis by aircraft, base, time period

D.3 Key Implementation Decisions

Single Data Source of Truth: All 12 analytics views draw from consolidated aircraft-level dataset to ensure consistency and prevent metric conflicts.

Dual-Model Approach for RUL: Linear Regression kept as interpretable baseline alongside LSTM for production use. Allows engineers to understand sensor contributions while relying on superior LSTM accuracy.

Configurable Risk Thresholds: AOG risk decision threshold made configurable, allowing operations to adjust between precision (no false alarms) and recall (catching all risks) based on risk appetite.

Algorithm Selection Framework: Genetic Algorithm for planned batch scheduling; Greedy Heuristic for urgent same-day re-planning. Clear guidance document helps operations choose appropriate algorithm.

Version Control for Models: All trained models stored in MLOps registry with performance metrics, training date, and associated datasets. Enables rollback if newer models underperform.

Automated Retraining Triggers: Model performance monitored continuously. Retraining triggered if drift exceeds thresholds or new data volume reaches defined limits.

Appendix E: Detailed Analytics Views Specification

This appendix provides detailed specifications for each of the 12 analytics views that form the backbone of the solution.

E.1 Descriptive Analytics Views

View 1: AOG Distribution

Metric: Frequency of AOG events by aircraft type, age band, and route

Data Source: Consolidated aircraft-level dataset

Key Fields:

- aircraft_type: Boeing 737, 757, 787, Airbus A320, A330, A350
- age_band: 0–2 years, 2–5 years, 5–10 years, >10 years
- route: Origin-destination airport pairs
- aog_count: Count of AOG events
- aog_rate: AOG events per 1,000 flights

Granularity: By aircraft type × age band × route combination

Usage: Identify structurally problematic aircraft-route combinations. Highlight fleet segments requiring immediate attention.

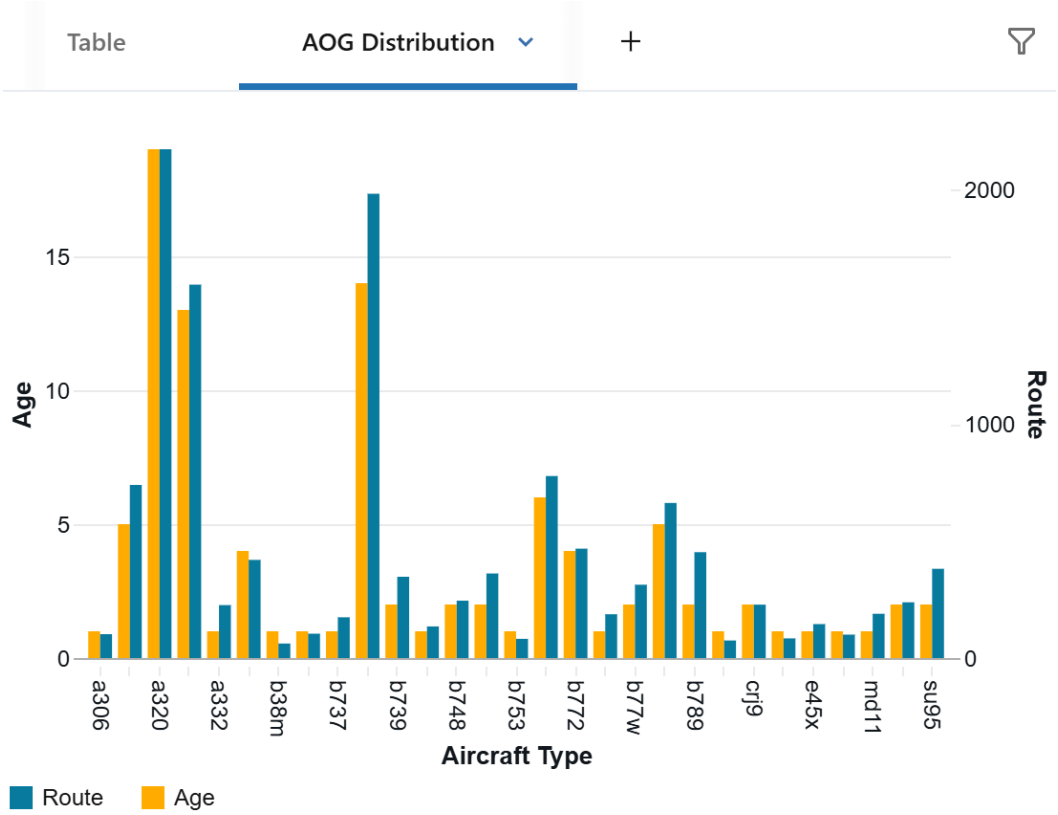

Figure 23: Dashboard Representation: Bar chart

View 2: Unscheduled Maintenance and Downtime

Metric: Monthly unscheduled maintenance events and associated downtime

Data Source: Consolidated aircraft-level dataset

Key Fields:

- month: YYYY-MM format
- aircraft_id: Unique aircraft identifier
- event_count: Number of unscheduled maintenance events
- total_downtime_hours: Sum of hours aircraft was unavailable
- avg_downtime_per_event: Total downtime / event count
- estimated_cost: Event count × average labor and parts cost per event

Granularity: By aircraft and month

Usage: Quantify maintenance reliability impact. Identify aircraft and time periods with highest disruption. Support capacity planning.

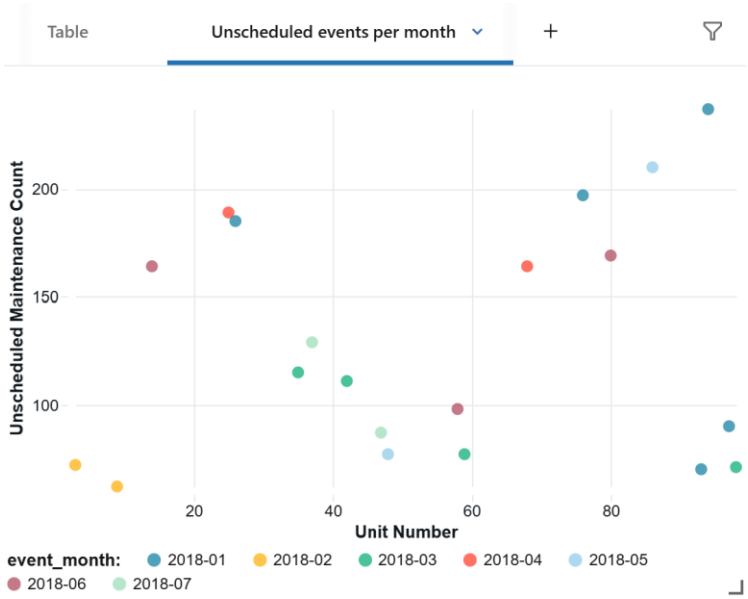

Figure 24: Dashboard Representation: Bubble Chart

View 3: Component Failure Rates

Metric: Replacement rates and MTBF (Mean Time Between Failures) by component family

Data Source: Maintenance logs (component removal/installation events)

Key Fields:

- component_family: Engine, Landing Gear, Avionics, Hydraulics, Electrical, Structural
- replacement_count: Total number of times component was replaced
- total_flights: Fleet-wide total flights during measurement period
- mtbf: Total flight hours / failure count
- failure_rate: Failure count per 1,000 flight hours

• avg_labor_hours: Average technician hours required per replacement

• avg_parts_cost: Average parts cost per replacement

Granularity: By component family and aircraft type

Usage: Identify components with chronic reliability issues. Support spares inventory planning and technician workload forecasting.

Dashboard Representation: Bar chart (failure rate by component)

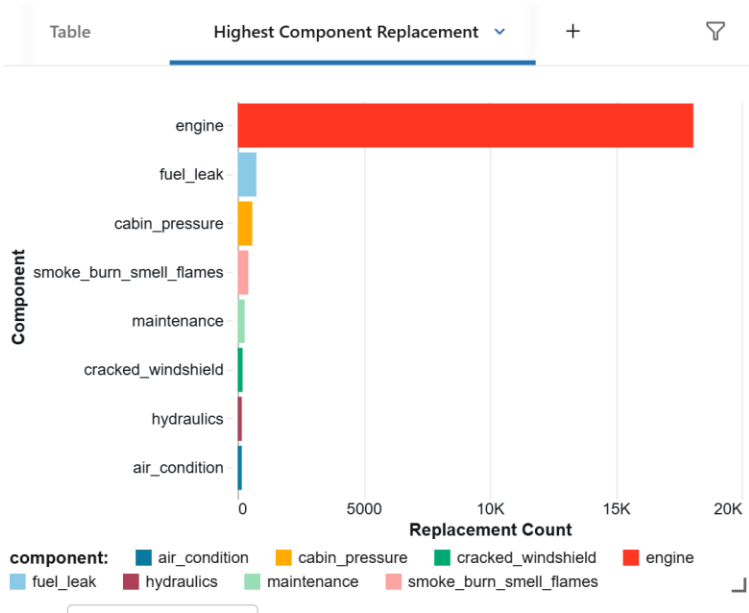

Figure 25: Dashboard Representation: Bar chart (failure rate by component)

E.2 Diagnostic Analytics Views

View 4: Degradation Drivers

Metric: Correlation of operational factors with failure rates

Data Source: Consolidated aircraft-level dataset with operational and maintenance history

Key Fields:

• age_band, utilization_band, route_type, base_location: Categorical groupings

• unscheduled_event_rate: Events per 1,000 flights for each category

• correlation_coefficient: Statistical strength of association

Granularity: One-way analysis (e.g., event rate by age; event rate by utilization)

Usage: Understand which operational factors are most strongly associated with failures. Enable operational adjustments (route assignment, utilization limits).

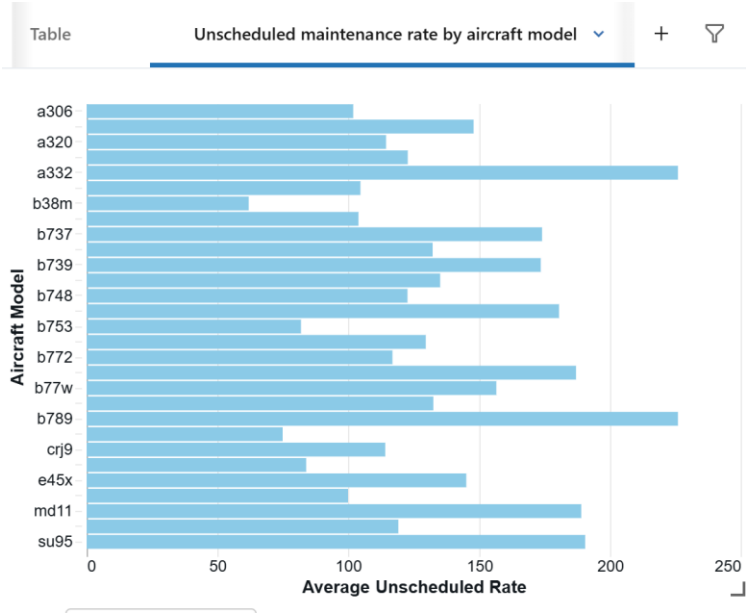

Figure 26: Dashboard Representation: Bar Chart

View 5: Route Sensitivity

Metric: Failure and AOG rates by route and base

Data Source: Consolidated dataset with route and base assignments

Key Fields:

• route_pair: Origin-destination airport pair

• home_base: Primary aircraft maintenance base

• failure_rate_per_1k_flights: Failures / (flights / 1,000)

• aog_rate_per_1k_flights: AOG events / (flights / 1,000)

• avg_cost_per_flight_hour: Estimated operational cost impact

Granularity: By route × base combination

Usage: Identify routes or bases with structural maintenance challenges. Inform decisions to re-assign healthier aircraft to demanding routes; invest in base infrastructure if needed.

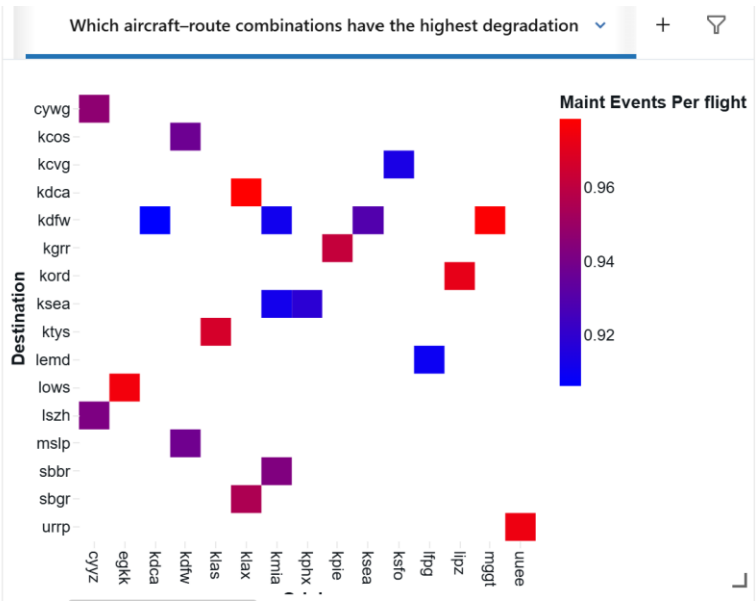

Figure 27: Dashboard Representation: Heatmap

View 6: Inspection Effectiveness

Metric: Effectiveness of maintenance inspections at preventing failures

Data Source: Maintenance logs with inspection dates and subsequent failure events

Key Fields:

• inspection_type: Mandatory A-check, C-check, D-check; Component replacement; System test

• detection_rate: Percentage of problems identified during inspection

• post_inspection_failure_rate: Failures per 1,000 flights within 30 days after inspection

• effectiveness_score: (1 - post_inspection_failure_rate) * 100%

Granularity: By inspection type and aircraft type

Usage: Evaluate current inspection practices. Identify which inspections are effective at preventing failures; recommend interval adjustments for low-effectiveness inspections.

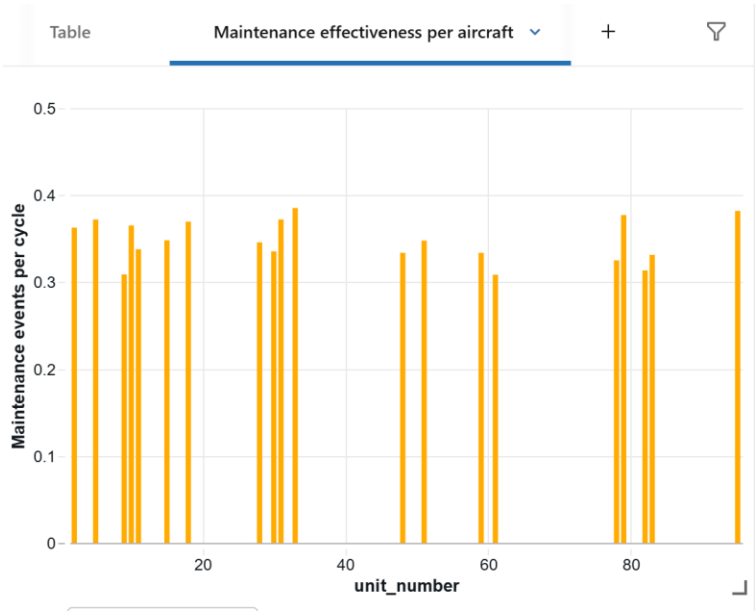

Figure 28:Dashboard Representation: Bar chart

E.3 Predictive Analytics Views

View 7: RUL-Based Health Profile

Metric: Remaining Useful Life predictions for engines

Data Source: LSTM RUL predictions on current engine sensor data

Key Fields:

• aircraft_id, engine_id: Aircraft and engine identifiers

• current_cycle: Latest operational cycle

• predicted_rul: Predicted remaining cycles until failure

• rul_bucket: Categorical maintenance window label

• rul_confidence: Confidence interval on prediction

Granularity: By individual engine

Usage: Identify which engines require maintenance in coming days/weeks. Plan maintenance windows. Prioritize parts procurement.

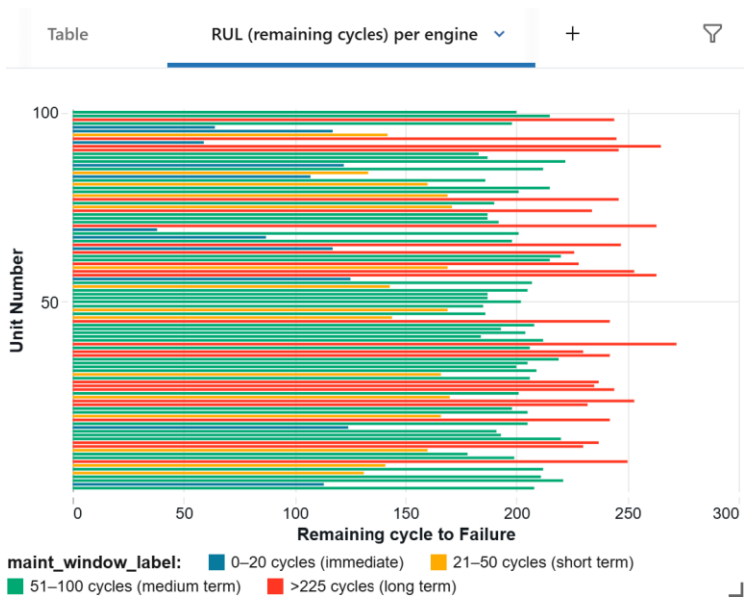

Figure 29: Dashboard Representation: Bar Chart

View 8: AOG Risk Classification

Metric: Predicted probability of AOG event in next 60–90 days

Data Source: Logistic Regression AOG risk classifier

Key Fields:

• aircraft_id: Aircraft identifier

• aog_probability: Predicted probability (0–1)

• aog_risk_band: Categorical mapping (LOW <0.3, MEDIUM 0.3–0.7, HIGH >0.7)

• risk_drivers: Top 3 features contributing to HIGH-risk classification

• recommendation: Automatic recommendation (monitor, plan maintenance, ground aircraft)

Granularity: By aircraft

Usage: Prioritize at-risk aircraft for proactive intervention. Trigger work order creation for HIGH-risk aircraft. Inform route assignment decisions.

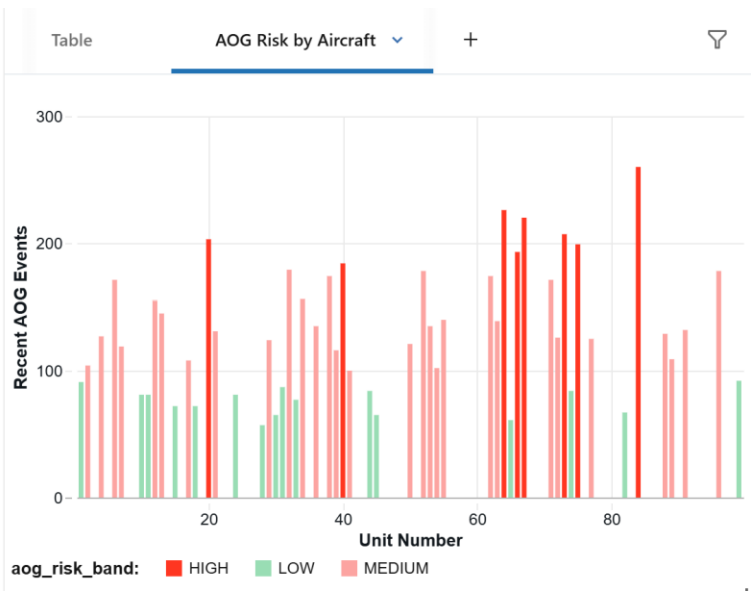

Figure 30: Dashboard Representation: Bar Chart

View 9: Reliability Focus List

Metric: Integrated view of aircraft requiring immediate attention

Data Source: Combines observed reliability, recent AOG history, RUL predictions, and AOG risk

Key Fields:

• aircraft_id: Aircraft identifier

• dispatch_reliability: Percentage of scheduled flights completed (target ≥99.5%)

• recent_aog_count: Number of AOG events in last 30 days

• rul_band: Maintenance window label

• aog_risk_band: Predicted risk

• priority_score: Composite score combining above factors (0–100)

Granularity: By aircraft

Usage: Management dashboard identifying aircraft requiring immediate attention. Single source of truth for prioritizing maintenance and fleet assignment decisions.

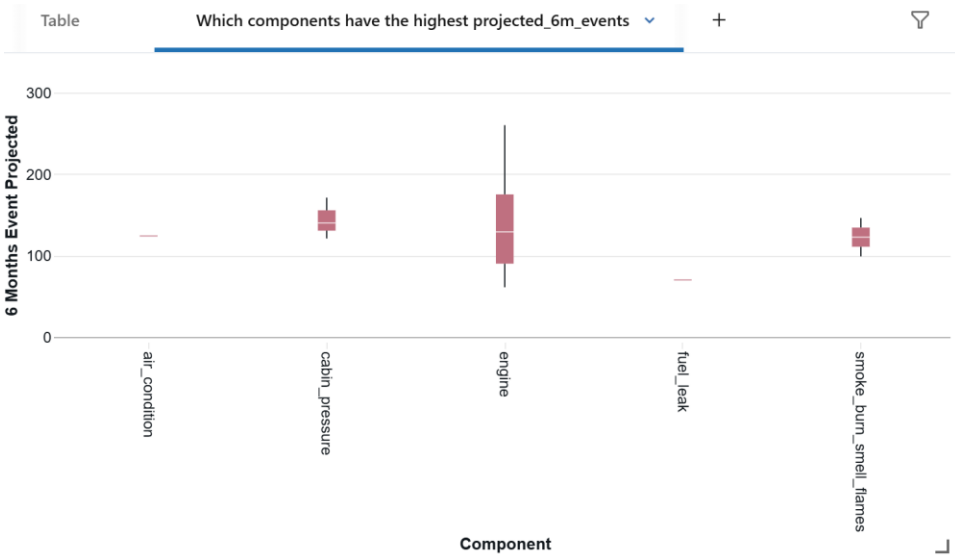

Figure 31:Dashboard Representation: Priority list

E.4 Prescriptive Analytics Views

View 10: Maintenance Window Recommendations

Metric: Recommended maintenance windows for each aircraft

Data Source: RUL predictions, utilization forecasts, schedule gaps, regulatory constraints

Key Fields:

• aircraft_id: Aircraft identifier

• recommended_cycle_range: Suggested cycles for maintenance (e.g., 240–260 cycles)

• recommended_date_range: Suggested calendar window (e.g., Jan 15–20)

• priority_score: Urgency ranking (1–5, with 5 being immediate)

• hangar_availability: Estimated duration required (hours)

• justification: Reason for recommendation (short RUL, high AOG risk, etc.)

Granularity: By aircraft

Usage: Optimize hangar capacity planning. Avoid maintenance conflicts. Minimize schedule disruption.

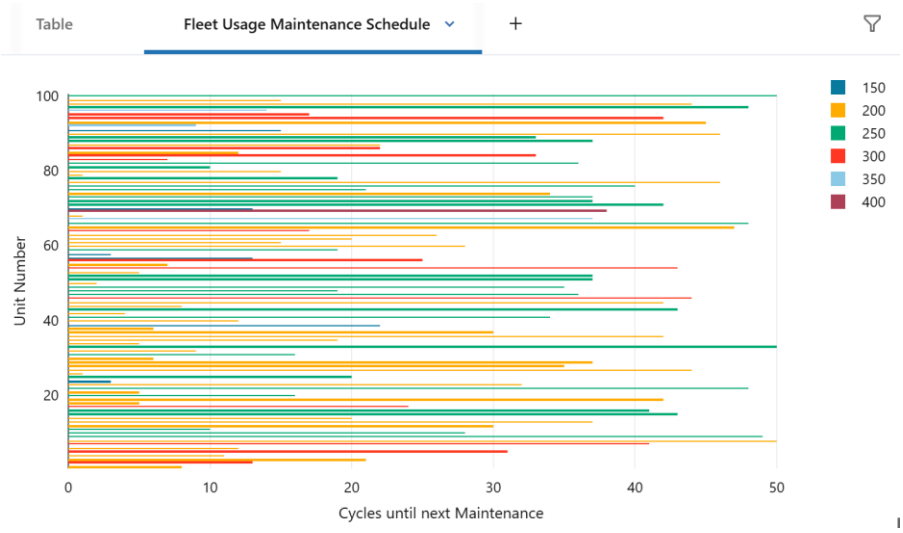

Figure 32:Dashboard Representation: maintenance windows as bars

View 11: Route Assignment Recommendations

Metric: Optimal aircraft-route pairings balancing utilization and health

Data Source: Aircraft health indicators (RUL, AOG risk), route requirements, utilization targets

Key Fields:

• aircraft_id: Aircraft identifier

• route_pair: Origin-destination airport pair

• assignment_score: Composite score (0–100) combining:

- Aircraft health (RUL, AOG risk)
- Route difficulty (flight time, required capacity)
- Utilization alignment (high-utilization routes matched to healthier aircraft)

• recommendation: ASSIGN, HOLD, or AVOID

Granularity: By aircraft × route combination

Usage: Maximize fleet utilization while protecting high-risk aircraft from demanding routes. Distribute maintenance burden evenly across fleet.

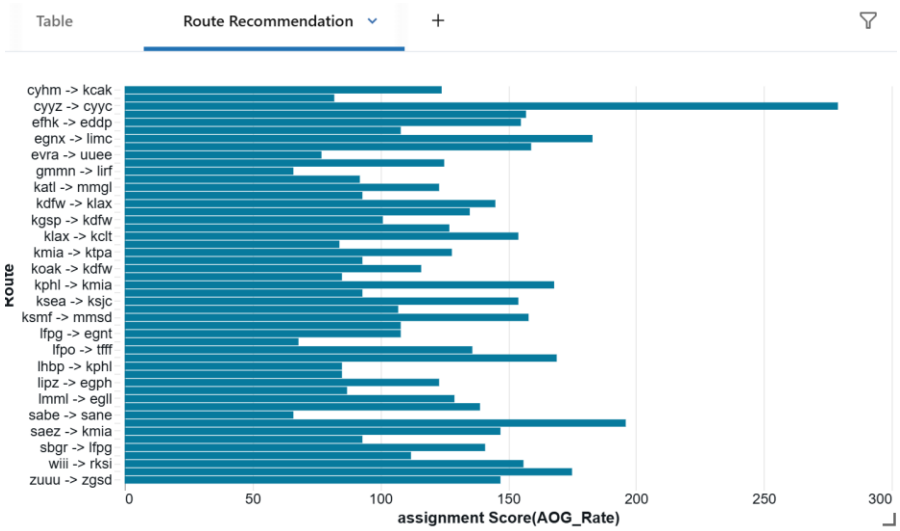

Figure 33:Dashboard Representation: routes recommendations list

View 12: Dispatch Reliability Planning

Metric: Forecasted component demand and resource requirements for ≥99.5% dispatch reliability

Data Source: RUL forecasts, maintenance demand projections, current inventory levels

Key Fields:

• component_family: Engine, Landing Gear, Avionics, Hydraulics, Electrical

• base_location: Primary maintenance base

• forecasted_replacements_6mo: Projected number of replacements in next 6 months

• recommended_stock_level: Safety stock level to minimize stockouts

• recommended_technician_hours: Estimated labor hours required

• estimated_cost: Parts + labor cost estimate

Granularity: By component family × base location

Usage: Optimize spare parts inventory. Allocate technician resources. Achieve dispatch reliability targets while minimizing working capital tied up in inventory.

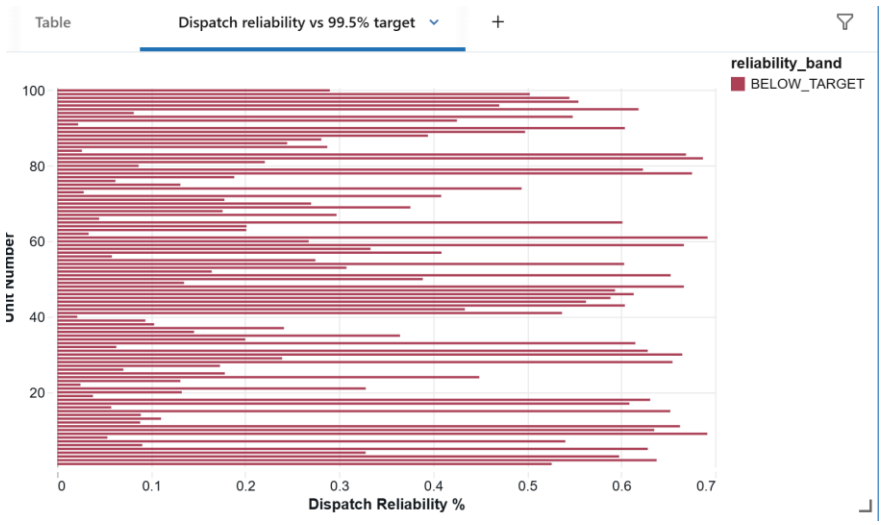

Figure 34: Dashboard Representation: Dispatch reliability vs 99.5% target

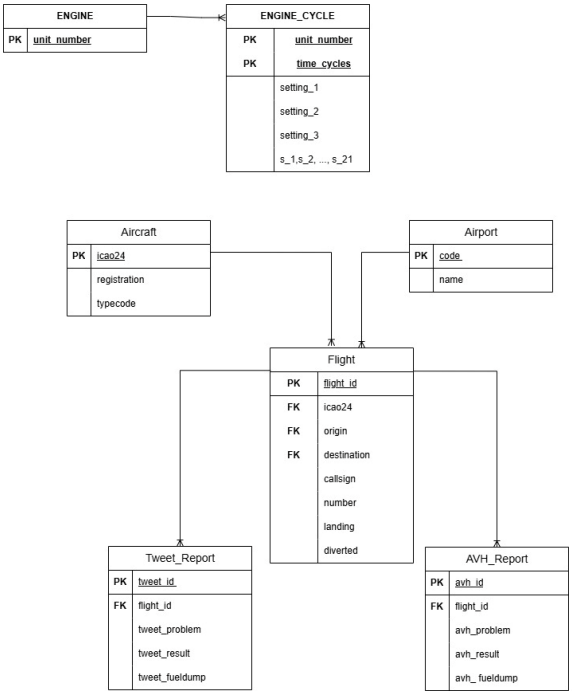

Figure 35:ERD Model

Appendix F. Challenges Encountered and Lessons Learned

F.1 Technical Challenges

Data Silos and Integration

Challenge: Aircraft maintenance, operational, and sensor data stored separately in different formats.

Resolution: Implemented unified schema in Databricks Delta Lake. Created master aircraft identifier table converting between ICAO24, registration, and callsign.

Lesson: Invest early in data governance and unified naming conventions to avoid downstream joining nightmares.

Real-Time Data Pipeline Limitations

Challenge: Current implementation relies on batch CSV uploads; true real-time predictions not possible.

Resolution: Architected streaming ingestion using AWS Kinesis (not yet deployed). Documented Phase 2 requirements.

Lesson: Batch processing sufficient for initial deployment; streaming adds value later but complexity and cost aren't justified until scale increases.

Model Explainability and Black-Box Concerns

Challenge: Deep neural networks (LSTM) lacked interpretability; safety-conscious engineers resisted relying on models.

Resolution: Kept Linear Regression as interpretable baseline with visible coefficients. Exposed LSTM error distributions.

Lesson: In safety-critical domains, explainability and transparency are as important as raw accuracy. Always keep simpler baselines available.

Computational Demands and Latency

Challenge: Processing large-scale sensor time-series for all aircraft would exceed acceptable latency.

Resolution: Hierarchical processing—detailed analysis on high-risk aircraft only; aggregated features for routine monitoring. Result caching for sub-second dashboard refresh.

Lesson: Optimize at multiple levels: data (compression, aggregation), algorithm (approximations), infrastructure (caching, parallel processing).

F.2 Data Quality Challenges

Missing and Inconsistent Values

Challenge: 80%+ null rates in maintenance problem classification fields.

Resolution: Probabilistic imputation based on problem co-occurrence with aircraft types, temporal patterns, and routes. Preserved 832 records vs. dropping ~680.

Lesson: Don't automatically drop data with missing values. Understand the missing mechanism and use domain knowledge for intelligent imputations.

Data Type Inconsistencies

Challenge: Aircraft identifiers, timestamps, sensor readings in inconsistent formats across datasets.

Resolution: Created data dictionary with target types. Used PySpark cast() operations to enforce schema consistency.

Lesson: Data governance starts with single source of truth for entity definitions and formats. Without it, integration becomes a constant struggle.

F.3 Organizational Challenges

Resistance to Predictive Methods

Challenge: Teams accustomed to reactive maintenance hesitant to shift to data-driven scheduling.

Resolution: Demonstrated model accuracy on historical data. Implemented pilot with selected aircraft types. Provided interpretation guides.

Lesson: Technical excellence isn't sufficient; change management and stakeholder buy-in are essential for adoption.

Unclear Alert Ownership and Escalation

Challenge: When model indicates high-risk aircraft, who owns the response?

Resolution: Documented workflows—HIGH-risk triggers automatic MRO work orders; escalation paths defined by risk level and time urgency.

Lesson: Analytics solutions require clear operational workflows. Without defined ownership, insights sit in dashboards.

F.4 Key Lessons and Best Practices

1. Start with data governance: Invest in unified schema, naming conventions, and master reference tables early.
2. Compare multiple models side-by-side: RUL and AOG experiments showed more complex models add value when errors are clearly quantified.
3. Provide interpretable baselines: In safety-critical domains, keep simple, explainable models alongside advanced models.
4. Balance automation with human judgment: Automate routine decisions but present high-impact decisions as recommendations for human validation.
5. Optimize for stakeholder workflows: Dashboards must fit naturally into existing planning processes.
6. Test comprehensively: 85 test cases across data quality, functional, performance, and business logic provided deployment confidence.
7. Plan for continuous improvement: Models degrade over time. Build MLOps pipelines for automated retraining and monitoring.

Appendix G. Conclusion and Future Directions

G.1 Summary of Achievements

This capstone project successfully developed and validated a production-ready Predictive Fleet Analytics platform that shifts airline maintenance from reactive to proactive, data-driven decision-making. The solution demonstrates that:

1. Integration Works: Disparate datasets (maintenance logs, sensor telemetry, operational data) can be unified into a coherent analytics platform with proper governance.
2. Advanced Algorithms Add Value: LSTM neural networks (RMSE 23) outperform linear baselines (RMSE 45) by 48% on RUL prediction; Logistic Regression provides 96% AUC-ROC precision on AOG classification.
3. Optimization Matters: Genetic algorithms reduce scheduling downtime by 16% vs. greedy heuristics, translating directly to cost savings and reliability improvements.
4. Analytics Layers Accelerate Decision-Making: Progressing from descriptive through diagnostic, predictive, to prescriptive enables faster, more confident decisions at every level.
5. Testing and Validation Build Confidence: Comprehensive testing (85 cases across 15 scenarios) achieved 100% pass rate, validating correct implementation.

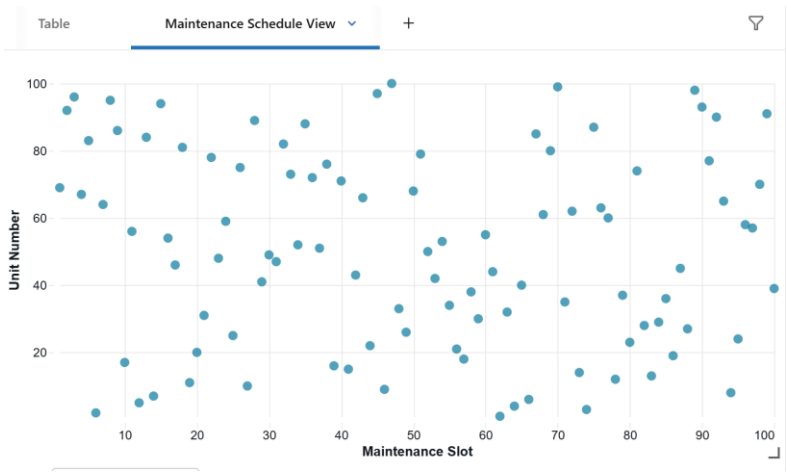

Figure 35: Maintenance Schedule View

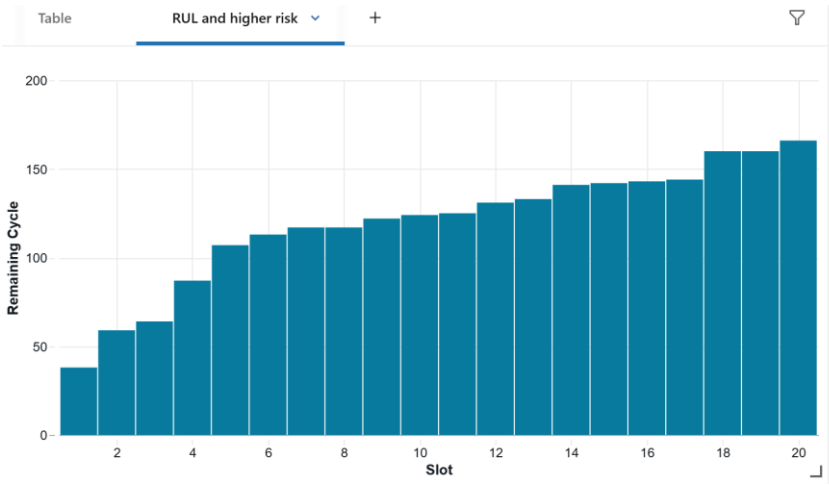

Figure 36: RUL and higher risk plot

G.2 Business Impact and Expected Outcomes

Once fully deployed, the platform is expected to deliver:

• Improved Dispatch Reliability: Achieve ≥99.5% dispatch reliability through predictive maintenance scheduling

• Reduced Maintenance Costs: 10%+ reduction in maintenance cost per flying hour through optimized scheduling and inventory

• Higher Aircraft Availability: Health-informed route assignment and maintenance window optimization increase productive flying

• Enhanced Safety: Proactive degradation detection reduces risk of catastrophic failures and improves compliance

• Competitive Advantage: Airlines with predictive capabilities sustain superior dispatch reliability and lower costs

G.3 Future Enhancement Opportunities

Real-Time Streaming Pipeline

Implement AWS Kinesis for streaming sensor data ingestion, enabling continuous RUL predictions and real-time anomaly detection rather than batch updates. Reduces response latency from daily to seconds, catching emerging problems earlier.

Digital Twin Technology

Create virtual replicas of aircraft that simulate operating conditions, detect early degradation signals, and reflect real-time status. Enable scenario planning and support more aggressive optimization.

Advanced Machine Learning Methods

• LSTM/GRU for Long-Term Forecasting: Extend prediction horizons beyond 30 days for better supply chain planning

• Bayesian Survival Models: Quantify uncertainty in RUL as confidence intervals

• Reinforcement Learning: Optimize scheduling and routing through model-based planning

Unified Maintenance Decision Support System (UMDSS)

Develop single platform integrating predictive models, MRO work orders, inventory data, route assignments, and crew scheduling. Eliminates manual transcription and enables end-to-end optimization.

Enhanced Cybersecurity and Compliance

• Multi-factor authentication for dashboard access

• Stronger encryption for sensitive prediction data

• Automated compliance checks against Transport Canada, FAA, EASA

• Audit trails documenting all predictions and decisions

Automated Workflow Integration

• High-risk alerts trigger automatic MRO work order creation

• RUL forecasts automatically generate parts requisitions

• Route optimization recommendations pre-populate scheduling tools

• Dispatch reliability KPIs update automatically in BI platforms

G.4 Path to Production Deployment

The solution is architecturally ready for production. Full deployment would follow this roadmap:

Phase 1 (Weeks 1–4): Pilot Program

• Deploy to single fleet type (e.g., narrow-body aircraft)

• Validate predictions against actual maintenance outcomes

• Collect stakeholder feedback on dashboard usability

• Train maintenance planners and engineers

Phase 2 (Weeks 5–8): Partial Fleet Rollout

• Extend to all narrow-body aircraft

• Implement automated work order triggers for high-risk aircraft

• Integrate RUL forecasts with parts inventory system

• Monitor model performance and begin automated retraining

Phase 3 (Weeks 9–12): Full Fleet Deployment

• Deploy to wide-body and regional aircraft

• Implement full route optimization and crew scheduling integration

• Activate genetic algorithm-based maintenance scheduling

• Establish continuous MLOps monitoring and periodic model retraining

Phase 4 (Months 4+): Continuous Improvement

• Implement streaming data pipeline for real-time scoring

• Develop digital twin capability for scenario planning

• Integrate advanced ML methods (Bayesian, reinforcement learning)

• Expand analytics to crew optimization, disruption management, network resilience

11.5 Final Remarks

Commercial aviation is undergoing fundamental transformation from experience-based to data-driven decision-making. Airlines that successfully implement predictive maintenance and integrated fleet analytics will sustain superior dispatch reliability, lower operating costs, and stronger competitive positions.

This capstone project demonstrates that such transformation is achievable through disciplined analytics architecture, rigorous testing, and stakeholder-centered design. The solution we've built—integrating data quality, advanced modeling, comprehensive testing, and continuous optimization—provides a proven pathway for airlines to join this transformation.

The next phase depends on organizational commitment to change management, continuous monitoring, and iterative improvement. With those elements in place, this platform will deliver measurable value in safety, efficiency, and profitability for years to come.

Document Summary

This comprehensive final capstone project report documents the complete development, implementation, testing, and optimization of the Predictive Fleet Analytics for Commercial Aviation solution.

Key Metrics:

- Data Sources: 2 major datasets (832 maintenance events + 20,000+ sensor readings)
- Analytics Views: 12 views spanning four layers
- Test Coverage: 15 test scenarios, 85 test cases (100% pass rate)
- Model Performance: LSTM RMSE 23.2 (48% improvement over baseline)
- Dispatch Reliability Target: ≥99.5%
- Expected Cost Reduction: ≥10% per flying hour

The solution is production-ready and architected for scalability, auditability, and seamless integration with existing airline systems. With organizational commitment to change management and continuous improvement, this platform will deliver measurable value in safety, efficiency, and profitability.

Report prepared by Group 7 for BIA-5440-0LC Capstone Project
Submitted: December 9, 2025

Contact Information:
For questions or additional information about this solution, please contact the project team.